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### Unlocking sharing destinations in an interaction system

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#### Abstract

A third-party user input content item is presented. A content sharing function is invoked responsive to determining user selection of a content sharing graphical element. The content sharing function comprises presentation of a destination graphical element identifying a first content sharing destination. The first content sharing destination is locked. A combination graphical element is user-selectable to invoke a combination function. Responsive to determining user selection of the combination graphical element, the combination function is invoked to access a second user input content item and combine the third-party user input content item with the second user input content item to create a combined user input content item. The first content sharing destination is unlocked and the user is enabled to share the combined user input content item to the unlocked first content sharing destination.

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## **Background/Summary**

CLAIM OF PRIORITY (1) This application claims the benefit of priority to U.S. Provisional Application Ser. No. 63/365,167, filed on May 23, 2022, U.S. Provisional Application Ser. No. 63/365,648, filed on Jun. 1, 2022, U.S. Provisional Application Ser. No. 63/353,818, filed Jun. 20, 2022, and U.S. Provisional Application Ser. No. 63/369,363, filed Jul. 25, 2022, each of which is incorporated herein by reference in its entirety.

### **BACKGROUND**

(1) The popularity of computer-implemented programs that permit users to access and interact with content and other users online continues to grow. Various computer-implemented applications exist that permit users to share content with other users through messaging clients. Some of such computer-implemented applications can be designed to run on a mobile device such as a phone, a tablet, or a wearable device, while having a backend service provided on a server computer system to perform certain operations, e.g., those that may require resources greater than is reasonable to perform at a user device.

(2) A media application such as a messaging client executing at a user device may provide a user interface that allows a user to capture a photo or a video and to share captured or viewed content to other devices. The media application may also provide a user interface that allows the user to modify or augment content before sharing the content to other devices.

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## **Description**

### **BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS**

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar

components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some non-limiting examples are illustrated in the figures of the accompanying drawings in which:

- (2) FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, according to some examples.
- (3) FIG. 2 is a diagrammatic representation of an interaction system, according to some examples, that has both client-side and server-side functionality.
- (4) FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, according to some examples.
- (5) FIG. 4 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (6) FIG. 5 is a flow diagram illustrating a method including aspects relating to a combination function, in accordance with some examples.
- (7) FIG. 6 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (8) FIG. 7 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (9) FIG. 8 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (10) FIG. 9 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (11) FIG. 10 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (12) FIG. 11 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (13) FIG. 12 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (14) FIG. 13 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (15) FIG. 14 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (16) FIG. 15 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (17) FIG. 16 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (18) FIG. 17 is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (19) FIG. 18 is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (20) FIG. 19 is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (21) FIG. 20 is a flow diagram illustrating a method including aspects relating to augmentation of user input content items, in accordance with some examples.
- (22) FIG. 21 is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (23) FIG. 22 is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (24) FIG. 23 is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.

- (25) FIG. **24** is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (26) FIG. **25** is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (27) FIG. **26** is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (28) FIG. **27** is a user interface diagram illustrating aspects relating to augmentation of user input content items in accordance with some examples.
- (29) FIG. **28** is a user interface sequence diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (30) FIG. **29** is a flow diagram illustrating a method including aspects relating to a combination function of a media application, in accordance with some examples.
- (31) FIG. **30** is a user interface sequence diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (32) FIG. **31** is a flow diagram illustrating a method including aspects relating to a combination function of a media application, in accordance with some examples.
- (33) FIG. **32** is a user interface sequence diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (34) FIG. **33** is a flow diagram illustrating a method including aspects relating to a combination function, in accordance with some examples.
- (35) FIG. **34** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (36) FIG. **35** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (37) FIG. **36** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (38) FIG. **37** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (39) FIG. **38** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (40) FIG. **39** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (41) FIG. **40** is a user interface diagram illustrating aspects relating to combined user input content items in accordance with some examples.
- (42) FIG. **41** is a user interface sequence diagram illustrating aspects relating to combined user input content items and unlocking of content sharing destinations in accordance with some examples.
- (43) FIG. **42** is a flow diagram illustrating a method including aspects relating to a combination function and unlocking of content sharing destinations, in accordance with some examples.
- (44) FIG. **43** is a flow diagram illustrating a method including aspects relating to a combination function and unlocking of content sharing destinations, in accordance with some examples.
- (45) FIG. **44** is a user interface diagram illustrating aspects relating to combined user input content items and unlocking of content sharing destinations in accordance with some examples.
- (46) FIG. **45** is a user interface diagram illustrating aspects relating to combined user input content items and unlocking of content sharing destinations in accordance with some examples.
- (47) FIG. **46** is a user interface diagram illustrating aspects relating to combined user input content items and unlocking of content sharing destinations in accordance with some examples.
- (48) FIG. **47** is a user interface diagram illustrating aspects relating to combined user input content items and unlocking of content sharing destinations in accordance with some examples.
- (49) FIG. **48** is an interaction diagram illustrating a process in accordance with some examples,



including aspects relating to sharing of combined user input content items to content collections.

(50) FIG. **49** is a flow diagram illustrating a method including aspects relating to sharing of combined user input content items to content collections.

(51) FIG. **50** is a flow diagram illustrating a method including aspects relating to sharing of combined user input content items to content collections.

(52) FIG. **51** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(53) FIG. **52** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(54) FIG. **53** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(55) FIG. **54** is a user interface diagram, showing part of a user interface, illustrating aspects relating to sharing of combined user input content items to content collections.

(56) FIG. **55** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(57) FIG. **56** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(58) FIG. **57** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(59) FIG. **58** is a user interface diagram illustrating aspects relating to sharing of combined user input content items to content collections.

(60) FIG. **59** is a user interface diagram, showing part of a user interface, illustrating aspects relating to sharing of combined user input content items to content collections.

(61) FIG. **60** is a diagrammatic representation of a message, according to some examples.

(62) FIG. **61** is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, according to some examples.

(63) FIG. **62** illustrates a system in which the head-wearable apparatus, according to some examples.

(64) FIG. **63** is a block diagram showing a software architecture within which examples may be implemented.

(65) Examples of the present disclosure improve the functionality of electronic software and systems by enabling users to modify, augment and combine content items to create combined content items in a media application. In examples of the present disclosure, users are enabled to combine multiple user input content items (e.g., multiple user-generated content items, referred to as UGCs) into combined user input content items, thereby providing enhanced software and system functionality and providing users with a powerful storytelling mechanism.

(66) Networked Computing Environment

(67) FIG. **1** is a block diagram showing an example interaction system **100** for facilitating interactions (e.g., exchanging text messages, conducting text audio and video calls, or playing games) over a network. The interaction system **100** includes multiple user systems **104**, each of which hosts multiple applications, including an interaction client **106** and other applications **108**. Each interaction client **106** is communicatively coupled, via one or more communication networks, including a network **110** (e.g., the Internet), to other instances of the interaction client **106** (e.g., hosted on respective other user systems **104**, an interaction server system **102** and third-party servers **112**). An interaction client **106** can also communicate with locally hosted applications **108** using Applications Program Interfaces (APIs).

(68) Each user system **104** may include multiple user devices, such as a mobile device **114**, head-wearable apparatus **116**, and a computer device **118**.

(69) An interaction client **106** interacts with other interaction clients **106** and with the interaction

server system **102** via the network **110**. The data exchanged between the interaction clients **106** (e.g., interactions **120**) and between the interaction clients **106** and the interaction server system **102** includes functions (e.g., commands to invoke functions) and payload data (e.g., text, audio, video, or other multimedia data).

(70) The interaction server system **102** provides server-side functionality via the network **110** to the interaction clients **106**. While certain functions of the interaction system **100** are described herein as being performed by either an interaction client **106** or by the interaction server system **102**, the location of certain functionality either within the interaction client **106** or the interaction server system **102** may be a design choice. For example, it may be technically preferable to initially deploy particular technology and functionality within the interaction server system **102** but to later migrate this technology and functionality to the interaction client **106** where a user system **104** has sufficient processing capacity.

(71) The interaction system **100** supports various services and operations provided to the interaction clients **106**. Such operations include transmitting data to, receiving data from, and processing data generated by the interaction clients **106**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information. Data exchanges within the interaction server system **102** are invoked and controlled through functions available via user interfaces (UIs) of the interaction clients **106**.

(72) Turning now specifically to the interaction server system **102**, an Application Program Interface (API) server **122** is coupled to and provides programmatic interfaces to interaction servers **124**, making the functions of the interaction servers **124** accessible to interaction clients **106**, other applications **108** and third-party server **112**. The interaction servers **124** are communicatively coupled to a database server **126**, facilitating access to a database **128** that stores data associated with interactions processed by the interaction servers **124**. Similarly, a web server **130** is coupled to the interaction servers **124** and provides web-based interfaces to the interaction servers **124**. To this end, the web server **130** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

(73) The API server **122** receives and transmits interaction data (e.g., commands and message payloads) between the interaction servers **124** and the user systems **104** (and for example, interaction clients **106** and other application **108**) and the third-party server **112**. Specifically, the API server **122** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the interaction client **106** and other applications **108** to invoke functionality of the interaction servers **124**. The API server **122** exposes various functions supported by the interaction servers **124**, including account registration; login functionality; the sending of interaction data, via the interaction servers **124**, from a particular interaction client **106** to another interaction client **106**; the communication of media files (e.g., images or video) from an interaction client **106** to the interaction servers **124**; the settings of a collection of media data (e.g., a story); the retrieval of a list of friends of a user of a user system **104**; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph); the location of friends within a social graph; and opening an application event (e.g., relating to the interaction client **106**).

(74) The interaction servers **124** host multiple systems and subsystems, described below with reference to FIG. 2.

(75) Linked Applications

(76) Returning to the interaction client **106**, features and functions of an external resource (e.g., a linked application **108** or applet) are made available to a user via an interface of the interaction client **106**. In this context, “external” refers to the fact that the application **108** or applet is external to the interaction client **106**. The external resource is often provided by a third party but may also be provided by the creator or provider of the interaction client **106**. The interaction client **106** receives a user selection of an option to launch or access features of such an external resource. The

external resource may be the application **108** installed on the user system **104** (e.g., a “native app”), or a small-scale version of the application (e.g., an “applet”) that is hosted on the user system **104** or remote of the user system **104** (e.g., on third-party servers **112**). The small-scale version of the application includes a subset of features and functions of the application (e.g., the full-scale, native version of the application) and is implemented using a markup-language document. In some examples, the small-scale version of the application (e.g., an “applet”) is a web-based, markup-language version of the application and is embedded in the interaction client **106**. In addition to using markup-language documents (e.g., a \*.ml file), an applet may incorporate a scripting language (e.g., a \*.js file or a .json file) and a style sheet (e.g., a \*.ss file).

(77) In response to receiving a user selection of the option to launch or access features of the external resource, the interaction client **106** determines whether the selected external resource is a web-based external resource or a locally-installed application **108**. In some cases, applications **108** that are locally installed on the user system **104** can be launched independently of and separately from the interaction client **106**, such as by selecting an icon corresponding to the application **108** on a home screen of the user system **104**. Small-scale versions of such applications can be launched or accessed via the interaction client **106** and, in some examples, no or limited portions of the small-scale application can be accessed outside of the interaction client **106**. The small-scale application can be launched by the interaction client **106** receiving, from a third-party server **112**, for example, a markup-language document associated with the small-scale application and processing such a document.

(78) In response to determining that the external resource is a locally-installed application **108**, the interaction client **106** instructs the user system **104** to launch the external resource by executing locally-stored code corresponding to the external resource. In response to determining that the external resource is a web-based resource, the interaction client **106** communicates with the third-party servers **112** (for example) to obtain a markup-language document corresponding to the selected external resource. The interaction client **106** then processes the obtained markup-language document to present the web-based external resource within a user interface of the interaction client **106**.

(79) The interaction client **106** can notify a user of the user system **104**, or other users related to such a user (e.g., “friends”), of activity taking place in one or more external resources. For example, the interaction client **106** can provide participants in a conversation (e.g., a chat session) in the interaction client **106** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using respective interaction clients **106**, with the ability to share an item, status, state, or location in an external resource in a chat session with one or more members of a group of users. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the interaction client **106**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

(80) The interaction client **106** can present a list of the available external resources (e.g., applications **108** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the application **108** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

(81) System Architecture

(82) FIG. 2 is a block diagram illustrating further details regarding the interaction system **100**,

according to some examples. Specifically, the interaction system **100** is shown to comprise the interaction client **106** and the interaction servers **124**. The interaction system **100** embodies multiple subsystems, which are supported on the client-side by the interaction client **106** and on the server-side by the interaction servers **124**. Example subsystems are discussed below.

(83) An image processing system **202** provides various functions that enable a user to capture and augment (e.g., annotate or otherwise modify or edit) media content associated with a message.

(84) A camera system **204** includes control software (e.g., in a camera application) that interacts with and controls hardware camera hardware (e.g., directly or via operating system controls) of the user system **104** to modify and augment real-time images captured and displayed via the interaction client **106**. The camera system **204** includes a number of subsystems, including a remix system **206** that enables users of the interaction system **100** to “remix” multiple content items to create a combined content item. For example, using the remix system **206**, a user, when viewing a collection of user input content items of one or more users of the interaction system **100**, can combine a user input content item with a selected further user input content item (e.g., image or video captured by the user, or an image or video selected by the user from a media library or file storage) to create a combined user input content item. The remix system **206** provides various tools to enable a user to creatively combine content items, such tools including a layout tool using which a relative layout of a first user input content item (or multiple user input content items) and a user input content item can be defined and specified.

(85) The augmentation system **208** provides functions related to the generation and publishing of augmentations (e.g., media overlays) for images captured in real-time by cameras of the user system **104** or retrieved from memory of the user system **104**. For example, the augmentation system **208** operatively selects, presents, and displays media overlays (e.g., an image filter or an image lens) to the interaction client **106** for the augmentation of real-time images received via the camera system **204** or stored images retrieved from memory **6202** of a user system **104**. These augmentations are selected by the augmentation system **208** and presented to a user of an interaction client **106**, based on a number of inputs and data, such as for example: Geolocation of the user system **104**; and Social network information of the user of the user system **104**.

(86) An augmentation may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo or video) at user system **104** for communication in a message, or applied to video content, such as a video content stream or feed transmitted from an interaction client **106**. As such, the image processing system **202** may interact with, and support, the various subsystems of the communication system **210**, such as the messaging system **212** and the video communication system **214**.

(87) A media overlay may include text or image data that can be overlaid on top of a photograph taken by the user system **104** or a video stream produced by the user system **104**. In some examples, the media overlay may be a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In further examples, the image processing system **202** uses the geolocation of the user system **104** to identify a media overlay that includes the name of a merchant at the geolocation of the user system **104**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the databases **128** and accessed through the database server **126**.

(88) The image processing system **202** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The image processing system **202** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

(89) The augmentation creation system **216** supports augmented reality developer platforms and

includes an application for content creators (e.g., artists and developers) to create and publish augmentations (e.g., augmented reality experiences) of the interaction client **106**. The augmentation creation system **216** provides a library of built-in features and tools to content creators including, for example custom shaders, tracking technology, and templates.

(90) In some examples, the augmentation creation system **216** provides a merchant-based publication platform that enables merchants to select a particular augmentation associated with a geolocation via a bidding process. For example, the augmentation creation system **216** associates a media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

(91) A communication system **210** is responsible for enabling and processing multiple forms of communication and interaction within the interaction system **100** and includes a messaging system **212**, an audio communication system **218**, and a video communication system **214**. The messaging system **212** is responsible for enforcing the temporary or time-limited access to content by the interaction clients **106**. The messaging system **212** incorporates multiple timers (e.g., within an ephemeral timer system **220**) that, based on duration and display parameters associated with a message or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the interaction client **106**. Further details regarding the operation of the ephemeral timer system **220** are provided below. The audio communication system **218** enables and supports audio communications (e.g., real-time audio chat) between multiple interaction clients **106**. Similarly, the video communication system **214** enables and supports video communications (e.g., real-time video chat) between multiple interaction clients **106**.

(92) A user management system **222** is operationally responsible for the management of user data and profiles, and includes a social network system **224** that maintains information regarding relationships between users of the interaction system **100**.

(93) A collection management system **226** is operationally responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **226** may also be responsible for publishing an icon that provides notification of a particular collection to the user interface of the interaction client **106**. The collection management system **226** includes a curation function that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **226** employs machine vision (or image recognition technology) and content rules to curate a content collection automatically. In certain examples, compensation may be paid to a user to include user-generated content into a collection. In such cases, the collection management system **226** operates to automatically make payments to such users to use their content.

(94) A map system **228** provides various geographic location functions and supports the presentation of map-based media content and messages by the interaction client **106**. For example, the map system **228** enables the display of user icons or avatars (e.g., stored in profile data **302**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the interaction system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the interaction client **106**. A user can furthermore share his or her location and status information (e.g., using an appropriate status

avatar) with other users of the interaction system **100** via the interaction client **106**, with this location and status information being similarly displayed within the context of a map interface of the interaction client **106** to selected users.

(95) A game system **230** provides various gaming functions within the context of the interaction client **106**. The interaction client **106** provides a game interface providing a list of available games that can be launched by a user within the context of the interaction client **106** and played with other users of the interaction system **100**. The interaction system **100** further enables a particular user to invite other users to participate in the play of a specific game by issuing invitations to such other users from the interaction client **106**. The interaction client **106** also supports audio, video, and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

(96) An external resource system **232** provides an interface for the interaction client **106** to communicate with remote servers (e.g., third-party servers **112**) to launch or access external resources, i.e., applications or applets. Each third-party server **112** hosts, for example, a markup language (e.g., HTML5) based application or a small-scale version of an application (e.g., game, utility, payment, or ride-sharing application). The interaction client **106** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the third-party servers **112** associated with the web-based resource. Applications hosted by third-party servers **112** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the interaction servers **124**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. The interaction servers **124** host a JavaScript library that provides a given external resource access to specific user data of the interaction client **106**. HTML5 is an example of technology for programming games, but applications and resources programmed based on other technologies can be used.

(97) To integrate the functions of the SDK into the web-based resource, the SDK is downloaded by the third-party server **112** from the interaction servers **124** or is otherwise received by the third-party server **112**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the interaction client **106** into the web-based resource.

(98) The SDK stored on the interaction system **100** effectively provides the bridge between an external resource (e.g., applications **108** or applets) and the interaction client **106**. This gives the user a seamless experience of communicating with other users on the interaction client **106** while also preserving the look and feel of the interaction client **106**. To bridge communications between an external resource and an interaction client **106**, the SDK facilitates communication between third-party servers **112** and the interaction client **106**. A WebViewJavaScriptBridge running on a user system **104** establishes two one-way communication channels between an external resource and the interaction client **106**. Messages are sent between the external resource and the interaction client **106** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

(99) By using the SDK, not all information from the interaction client **106** is shared with third-party servers **112**. The SDK limits which information is shared based on the needs of the external resource. Each third-party server **112** provides an HTML5 file corresponding to the web-based external resource to interaction servers **124**. The interaction servers **124** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the interaction client **106**. Once the user selects the visual representation or instructs the interaction client **106** through a GUI of the interaction client **106** to access features of the web-based external resource, the interaction client **106** obtains the HTML5 file and instantiates the resources to access the features of the web-based external resource.

(100) The interaction client **106** presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the interaction client **106** determines whether the launched external resource has been previously authorized to access user data of the interaction client **106**. In response to determining that the launched external resource has been previously authorized to access user data of the interaction client **106**, the interaction client **106** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the interaction client **106**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the interaction client **106** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the interaction client **106** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the interaction client **106**. The external resource is authorized by the interaction client **106** to access the user data under an OAuth 2 framework.

(101) The interaction client **106** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale applications (e.g., an application **108**) are provided with access to a first type of user data (e.g., two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of applications (e.g., web-based versions of applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

(102) An advertisement system **234** operationally enables the purchasing of advertisements by third parties for presentation to end-users via the interaction clients **106** and also handles the delivery and presentation of these advertisements.

(103) Data Architecture

(104) FIG. 3 is a schematic diagram illustrating data structures **300**, which may be stored in the database **304** of the interaction system **100**, according to certain examples. While the content of the database **304** is shown to comprise multiple tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

(105) The database **304** includes message data stored within a message table **306**. This message data includes, for any particular message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **306**, are described below with reference to FIG. 3.

(106) An entity table **308** stores entity data, and is linked (e.g., referentially) to an entity graph **310** and profile data **302**. Entities for which records are maintained within the entity table **308** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the interaction system **100** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

(107) The entity graph **310** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization), interest-based, or activity-based, merely for example.

(108) The profile data **302** stores multiple types of profile data about a particular entity. The profile

data **302** may be selectively used and presented to other users of the interaction system **100** based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **302** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the interaction system **100**, and on map interfaces displayed by interaction clients **106** to other users. The collection of avatar representations may include “status avatars,” which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

(109) Where the entity is a group, the profile data **302** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

(110) The database **304** also stores augmentation data, such as overlays or filters, in an augmentation table **312**. The augmentation data is associated with and applied to user input content, such as videos (for which data is stored in a video table **314**) and images (for which data is stored in an image table **316**).

(111) Filters, in some examples, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the interaction client **106** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the interaction client **106**, based on geolocation information determined by a Global Positioning System (GPS) unit of the user system **104**.

(112) Another type of filter is a data filter, which may be selectively presented to a sending user by the interaction client **106** based on other inputs or information gathered by the user system **104** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a user system **104**, or the current time.

(113) Other augmentation data that may be stored within the image table **316** includes augmented reality content items (e.g., corresponding to applying Lenses or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

(114) As described above, augmentation data includes augmented reality (AR), virtual reality (VR) and mixed reality (MR) content items, overlays, image transformations, images, and modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of the user system **104** and then displayed on a screen of the user system **104** with the modifications. This also includes modifications to stored content, such as video clips in a collection or group that may be modified. For example, in a user system **104** with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. Similarly, real-time video capture may use modifications to show how video images currently being captured by sensors of a user system **104** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

(115) Data and various systems using augmented reality content items or other such transform



systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In some examples, tracking of points on an object may be used to place an image or texture (which may be two-dimensional or three-dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

(116) Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

(117) In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of an object's elements, characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh.

(118) In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification, properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing the color of areas; removing some part of areas from the frames of the video stream; including new objects into areas that are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

(119) In some examples of a computer animation model to transform image data using face detection, the face is detected on an image using a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to determine facial feature reference points.

(120) Other methods and algorithms suitable for face detection can be used. For example, in some

examples, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

(121) A transformation system can capture an image or video stream on a client device (e.g., the user system **104**) and perform complex image manipulations locally on the user system **104** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the user system **104**.

(122) In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using the user system **104** having a neural network operating as part of an interaction client **106** operating on the user system **104**. The transformation system operating within the interaction client **106** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that are the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transform system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the user system **104** as soon as the image or video stream is captured and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured, and the selected modification icon remains toggled. Machine-taught neural networks may be used to enable such modifications.

(123) The graphical user interface, presenting the modification performed by the transform system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browsing to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

(124) A story table **318** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a

particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **308**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the interaction client **106** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

(125) A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the interaction client **106**, to contribute content to a particular live story. The live story may be identified to the user by the interaction client **106**, based on his or her location. The end result is a “live story” told from a community perspective.

(126) A further type of content collection is known as a “location story,” which enables a user whose user system **104** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end-user belongs to a specific organization or other entity (e.g., is a student on the university campus).

(127) As mentioned above, the video table **314** stores video data that, in some examples, is associated with messages for which records are maintained within the message table **306**. Similarly, the image table **316** stores image data associated with messages for which message data is stored in the entity table **308**. The entity table **308** may associate various augmentations from the augmentation table **312** with various images and videos stored in the image table **316** and the video table **314**.

(128) In some examples, if a user is viewing a user input content item, the user may wish to combine the user input content item with another content item. However, this may involve a number of manual steps or selections, and it may be desirable to reduce and/or automate these steps or selections, at least to some extent. One example technical problem of reducing and/or automating the above may be addressed by providing a reply user interface that automatically provides access to a combination function (also referred to as a “remix” function). In some examples, if a user is viewing a first user input content item, such as a third-party image, that allows users to combine the first user input content item with a second user input content item, such as a user-selected image, and reply to the first user input content item (e.g., “remix and reply”), a camera button is user-selectable to trigger an experience that includes the ability to access an augmentation function or tool, in the example form of a remix lens. The remix lens, also referred to as a combination lens or combination application element, allows the user to combine different user input content items to create a combined user input content item, which the user can then share and “reply” to the sharing of the first user input content item (e.g., the third-party image).

(129) FIG. 4 is a user-interface diagram depicting a user interface **404**, according to some examples, that may be presented by an interaction client **106**, in the example form of a media application, executing on a mobile device **114**. While the user interface **404** is described as being presented on the screen of a mobile device **114**, the user interface **404** may also be presented on the optical display of a head-wearable apparatus **116** or other wearable apparatus presenting a user interface, e.g., “smart contact lenses” or similar technology.

(130) The interaction system **100** allows users of the interaction system **100** to share and publish digital content, including various user input content items. User input content items may comprise user-selected content items (which may be user-selected content data), user-generated content items (which may be user-generated content data), computer-generated content items (which may be computer-generated content data, e.g., data generated by an artificial intelligence driven system), or

combinations thereof.

(131) The user interface **404** displays a user input content item in the example form of a third-party user input content item **406**, which, in some examples, may be part of a collection of third-party user input content items generated and published by a third-party user of the interaction system **100**. The third-party user input content item **406** may, for example, comprise image data, video data, audio data, or a combination of these data types, together with augmentation or modification data (e.g., stickers, filters or lenses).

(132) The third-party user input content item **406** may be part of a set of third-party user input content items created by a third-party user of the media application, and the presentation of the third-party user input content item may be part of a sequential presentation of the set of third-party user input content items.

(133) The user interface **404** also includes an interaction zone **402** with a function or tools section, in the example form of a function carousel **408**. The function carousel **408** presents various tools and functions that may be invoked by a user of the mobile device **114** to interact with other users of the interaction system **100** and also to interact with content made accessible via the interaction system **100**. The interaction zone **402** includes a combination graphical element **410** (e.g., a user-selectable icon) that is user-selectable to invoke a combination function that enables a user to combine the third-party user input content item **406** with another content item (e.g., an image or video). The user may combine the third-party user input content item **406** with another (second) user input content item chosen by the user, e.g., captured using a camera of the mobile device **114**, selected from local storage of the mobile device **114**, a cloud storage location, or another storage component, or partially or fully generated by a computer system or artificial intelligence. Various examples of the user interface **404** are discussed below. While certain examples of the present disclosure relate to the combination of a third-party user input content item with another content item, it should be appreciated that similar techniques may be applied to combine any user input content item (not necessarily being a third-party user input content item) with another content item.

(134) FIG. 5 is a flowchart illustrating a method **500**, according to some examples, to enable an active user to combine a third-party user input content item with another user input content item in a media application (e.g., the interaction client **106**). Accordingly, the method **500** may be performed by components of the interaction system **100**.

(135) The method **500** commences at opening loop block **502**, and progresses to block **504**, where the interaction client **106** causes the presentation of a third-party user input content item **406** within the user interface **404** of the mobile device **114**. The third-party user input content item **406** may be part of a set or collection of third-party user input content items published by a third-party user of the interaction system **100**. One or more third-party user may have created or generated one or more of these items and may have selected one or more of these items, e.g., from a local or cloud storage. In such examples, an active user of the interaction client **106** is enabled, within the context of the user interface **404**, to scroll in both backward and forward directions (or upward and downward directions) through the collection of third-party user input content items, or to automatically or manually advance display of the back of third-party user input content items.

(136) At block **506**, the remix system **206** of the interaction system **100** detects a third-party authorization for the third-party user input content item **406** to be combined with other content items in order to create a combined user input content item. For example, an authorization setting related to the third-party user input content item **406**, indicating this authorization, may be detected at block **506**. The determining third-party authorization for the third-party user input content item to be combined may thus comprise determining a user setting related to the third-party user input content item indicating the third-party authorization.

(137) A third-party content creator, when creating or publishing the third-party user input content item, is presented with the option of either allowing or disallowing the use of the third-party user input content item **406** in combination with other content items, created or originated with other

users of the interaction system **100**. This allowance/disallowance is then recorded and stored in the database **128** as the authorization setting related to the relevant third-party user input content item, and is accessed at block **506** to perform the detection of the third-party authorization.

(138) At block **508**, and responsive to determining the third-party authorization, the interaction client **106** automatically causes display or presentation, on the mobile device **114** of the user, of a combination graphical element. The automatic causing of the presentation of the combination graphical element includes causing presentation of the combination graphical element without user action, for example relative to the user interface **404**, as may be presented by the interaction client **106**. The combination graphical element is user-selectable to invoke a combination function provided by the remix system **206** of the interaction system **100**. The combination graphical element, in some examples, is a camera indicium (e.g., a camera button) that indicates that it is user-selectable to invoke the combination function. The combination graphical element is further caused to be displayed in conjunction and contemporaneously with the third-party user input content item **406** on a user interface **404** of the interaction client **106**.

(139) The combination function may provide a number of sub-functions or tools to facilitate the combining of the third-party user input content item **406** and the other user input content item. These tools may include a layout function to enable the user to specify a layout relationship of the third-party user input content item **406**, relative to the other user input content item, or within the context of a broader combined display, during the presentation of the combined user input content item. The combination function also makes a number of the augmentations of the augmentation system **208** discussed herein available for application and use in the creation of the combined content element.

(140) In some examples, the combination function, in addition to enabling the combining of multiple content items, also enables the triggering of a reply messaging function (e.g., provided by the messaging system **212**) to enable automatic messaging or sending of the combined user input content item to the third-party user of the interaction system **100**. The sending of the combined user input content item may be as a message (e.g., message **6000**) to the third-party user or by selective publication to a content feed of the third-party user. The reply messaging function may thus allow two users to collaborate on a content item together to create a combined content item. In some examples, this collaboration may occur in a private conversation. Accordingly, the user's reply to the transmission of the third-party user input content item may comprise transmitting the combined user input content item in a private conversation between the user and the third-party user facilitated by the interaction system **100**.

(141) At block **510**, user selection of the combination graphical element is detected and, at block **512**, the combination function provided by the remix system **206** is invoked responsive to determining the user selection. The combination function enables the active user to combine the third-party user input content item **406** with another user input content item, in order to create a new, combined user input content item. The method **500** then terminates at block **514**.

(142) While examples herein refer to combining digital content in the form of images and videos, it should be appreciated that audio items may also be combined with other content items (that may include image, video and/or audio content). For example, a viewing user may select a third-party user input content item comprising an audio item (e.g., a specific song linked to an image or video clip), and use the combination function to combine the audio item with another content item to create the combined user input content item (e.g., combine the specific song with another image that was captured or selected by the user).

(143) FIG. **6** is a user interface diagram showing a user interface **606**, according to some examples. Referring to the user interface **606**, where an active user is viewing a third-party user input content item in the example form of the third-party image **604** (another example may be a third-party video clip, or an image/video with added audio content) and responsive to determining a combination authorization with respect to the third-party image **604**, a combination graphical element in the

form of a camera button **602** is displayed. The camera button **602** is user-selectable to trigger the combination function (e.g., an experience) that includes the ability to access a combination augmentation or lens (e.g., a “remix” lens), in order to create a combined content element (e.g., a combined user input content item) that includes the third-party image **604**.

(144) FIG. 7 is a user interface diagram showing a user interface **702**, according to some examples. Referring to the user interface **702**, in some examples, where an active user is viewing the third-party image **604** described with reference to FIG. 6, and responsive to determining a combination authorization with respect to the third-party image **604**, a combination graphical element in the form of a remix button **704** is displayed.

(145) The remix button **704** is user-selectable to trigger the combination function. The remix button **704** differs from the camera button **602** in that it specifically includes a graphical component (e.g., the arrow shown in conjunction with the camera image in FIG. 7) that identifies the combination function, including its reply message capabilities. The graphical component may provide a contextual signal to the user, indicative of the option to trigger the combination function and its reply message capabilities. The arrow shown in FIG. 7 is merely an example, and it should be appreciated that examples of the present disclosure may include various identifiers of the combination function, e.g., various different augmentations or supplements to a camera button, to alert the active user of the option to activate the combination function.

(146) FIG. 8 is a user interface diagram showing a user interface **802**, according to some examples. When an active viewer is viewing the third-party image **604** within the user interface **802** for which a combination authorization is recorded and accessed, user selection of a camera button **602** opens a modular reply camera interface **904** (e.g., provided by the reply system of the camera system **204**), as shown in FIG. 9. Accordingly, invoking the combination function (by selecting the camera button **602**) may include activating and opening the modular reply camera interface.

(147) Referring to the user interface diagram in FIG. 9, the reply camera interface **904** presents an augmentation carousel **906**. Within the augmentation carousel **906**, a combination graphical element in the form of a remix lens **908**, also referred to as a combination lens or combination application element, is located at a first position relative to other lenses within the augmentation carousel **906**, and is preselected (e.g., as indicated by a visual highlighting and/or size differentiation and/or its central position in the reply camera interface **904**). The augmentation carousel **906** is automatically applied to merge or combine the third-party image **604** with a second user input content item, in this case an image captured by the active user, to create a combined image **902** within the reply camera interface **904**. Accordingly, in some examples, invoking the combination function causes the modular reply camera interface to be displayed, and includes preselecting, by the media application, the combination lens (combination application element), allowing the user convenient access to the remix function.

(148) FIG. 10 is a user interface diagram showing a viewing user interface **1002**, according to some examples. When an active user is viewing the third-party image **604**, for which a combination authorization is stored and accessed, within the viewing user interface **1002**, user selection of a camera button **602** activates and opens a modular reply camera interface **1104** (e.g., provided by the camera system **204**) as shown in FIG. 11.

(149) FIG. 11 is a user interface diagram showing the reply camera interface **1104**, according to some examples. The reply camera interface **1104** presents a user input content image **1102**, e.g., a “selfie” image captured by the active user (self-portrait photograph captured using a front camera, for example), which image may have been captured responsive to user selection of the camera button **602**. The reply camera interface **1104**, automatically and by default, presents an augmentation carousel **1106** within which a remix camera **1108** (or remix lens, also referred to as a combination lens) is a first lens in the augmentation carousel **1106** but, in contrast with the example discussed with reference to FIG. 8 and FIG. 9, is not preselected. The remix camera **1108** is referred to as the first lens in the augmentation carousel **1106**, because it is closest to the central

capture button **1112**. The remix camera **1108** is not preselected and the central capture button **1112**, which can be used to capture a standard image (without re-mixing or augmentation) is preselected and displayed larger than the icons in the augmentation carousel **1106**.

(150) The active user may then scroll the augmentation carousel **1106** to the left to move the remix camera **1108** into the central position (see FIG. **12**), or select the remix camera **1108** by pressing it directly, in order to activate it. Using the combination function of the remix camera **1108**, the active user can conveniently and easily combine the third-party image **604**, from a third-party user of the interaction system **100** (e.g., in this example, “Sally”) and the user input content image **1102** (e.g., a selfie captured in by the active user using the remix camera **1108** or selected from a camera roll of a mobile device), to create a combined user input content item. In this example, the combined user input content item is in the form of a combined image **1202** (see FIG. **12**, which is described below).

(151) FIG. **12** is a further user interface diagram showing the reply camera interface **1104** with the combined image **1202**. As shown in FIG. **12**, the reply camera interface **1104** may also include a set of layout graphic elements **1204**, each of which is user-selectable to apply a different layout to the combination of the third-party image **604** and the user input content image **1102**. Each of the layout options associated with a respective layout graphic element **1204** may allocate a certain section of an interface canvas to each of the images or may define a layout of the images relative to each other. The same layout options may be applied to, or available in respect of, still images and video content. (In some examples, a combined user input content item may include a still image combined with video content and various layout options may likewise be available.)

(152) The combined image **1202** in FIG. **12** utilizes a layout option in terms of which the active user's image is overlaid onto the third-party image **604** (the latter essentially becoming a background image). Another example may be a “divided screen remix,” in which the active user's image is shown in a first section or block, and the third-party image **604** is shown in a second section or block, separate from the first section or block within the interface.

(153) The reply camera interface **1104** also includes a reply graphic element **1110** that provides contextual information regarding the combination function (e.g., in this example, indicating that the active user is creating the combined image **1202** intended for a reply to the third-party user “Sally”). The reply graphic element **1110** may also be user-selectable to invoke the messaging system **212** of the interaction system **100** and conveniently enable the active user to transmit (e.g., by way of a message or publication to a feed) the combined image **1202** to the third party user (e.g., “Sally”) by way of a “reply” to the third party's publication of the third-party image **604**.

(154) The combination function provided by the media application may thus include a reply function to enable the user to transmit a combined user input content item to a third-party user of the media application by way of reply to a transmission of the third-party user input content item.

(155) FIG. **13** to FIG. **16** show user interface diagrams representing a series of user interfaces **1302** in sequence, according to some examples. Here, when a user selects a camera button **602** (FIG. **13**), the augmentation carousel **1502** is not automatically generated and caused to be presented within the user interface **1302**.

(156) Rather, a carousel icon **1402** is displayed adjacent to a central (and preselected) image capture graphical element **1404**, which is user-selectable to capture an image using a camera of the mobile device **114**. This carousel icon **1402** is shown in FIG. **14**.

(157) The carousel icon **1402** is user-selectable to generate display of an augmentation carousel **1502**, as shown in FIG. **15**, in which the remix camera **1504**, or combination graphical element, is a first content augmentation option (e.g., “lens”) presented within the sequence of augmentation options of the augmentation carousel **1502**.

(158) As shown in FIG. **14** to FIG. **16**, a reply graphic element **1406**, as described elsewhere, is provided in an upper region of the user interface **1302**.

(159) The user is also provided with various options in terms of laying out the two content items

relative to each other, accessible via the layout graphic elements **1602**, as described elsewhere, and displayed according to some examples in FIG. **16**.

(160) FIG. **17** is a user interface diagram showing a viewing user interface **1702**, according to some examples, that is presented when an active, viewing user of the interaction client **106** views a third-party image **604**. In this example, a tooltip **1704** is generated and caused to be displayed in conjunction with the remix button **1706** (in FIG. **17**, this is a supplemented camera button). The tooltip **1704** may be displayed and presented when a user taps on the remix button **1706** or when the user initially views the third-party image **604**. The tooltip **1704** may also be presented for a set time (e.g., between 2 and 10 seconds, and then disappear from the viewing user interface **1702**) and may be presented to a user a set number of times (e.g., up to a maximum of three times) within any interaction session with the interaction client.

(161) In some examples, if a user is viewing a user input content item such as a third-party image that has an augmentation applied to it (e.g., a lens, filter, or other augmentation applied to the third-party image), the user may wish to apply the same augmentation, but this may require a number of manual steps/selections. One example technical problem of automatically providing access to a camera function together with a content combination function and/or content augmentation function, e.g., an augmentation function allowing the viewing user to access and automatically apply the same augmentation as applied to a third-party image, may be addressed by a method and system according to examples of the present disclosure, which provides an improved reply camera function. A camera button is user-selectable to trigger a function (e.g., experience) that provides the ability for the viewing user conveniently to access and apply the same augmentation as applied to the third-party image to user input content items. According to some examples, a viewing user interface includes an interaction zone that provides a top-level camera button displayed concurrently with digital content, e.g., a user input content item, also displayed within the context of the viewing user interface. The camera button is supplemented or modified to indicate user selection of the camera button triggers a specific function (e.g., an “experience” such as a remix experience or a lens experience). The camera button may be supplemented or modified in various ways, including highlighting the camera button with an appropriate thumbnail image (e.g., a lens thumbnail image or a remix thumbnail image) associated with a function or user experience that is invoked responsive to selection of the modified camera button. Selection of the modified camera button invokes that associated function, in addition to opening of a modular reply camera. In other words, according to some examples, user selection of the modified camera button invokes both a camera function (e.g., a “reply” camera function which is a camera function within the broader context of a reply before applying to a user input content item posted by a third party) as well as a further function for use with the camera function (e.g., a content combination function or a content augmentation function).

(162) FIG. **18** is a user interface diagram depicting a viewing user interface **1802**, according to some examples, that may be presented by an interaction client **106**, in the example form of a media application, executing on the mobile device **114**. While the viewing user interface **1802** is described as being presented on the screen of the mobile device **114**, the viewing user interface **1802** may also be presented on the optical display of a head-wearable apparatus **116** or other wearable apparatus presenting a user interface, e.g., “smart contact lenses” or similar technology.

(163) The viewing user interface **1802** displays a third-party user input content item **1804**, which, in some examples, may be part of a collection of third-party user input content items generated and published by a third-party user of the interaction system **100**. The third-party user input content item **1804** may comprise image data, video data, audio data, or one or more content data comprising a combination of these data types, together with augmentation or modification data (e.g., stickers, filters or lenses).

(164) The viewing user interface **1802** includes an interaction zone **1806** that includes an augmentation camera graphical element **1808** (e.g., a user-selectable camera button **2102**) that is



user-selectable to invoke a reply camera function that enables a user to respond to the publication of the third-party user input content item **1804** by a third-party user. This response may include a user input content item (e.g., a digital image captured using the camera of the mobile device **114**). Additionally, the invoked reply camera function may enable the user to augment the user input content item that the user wishes to add or include with an augmentation that is applied to the third-party user input content item **1804**. To this end, user selection of the augmentation camera graphical element **1808** causes display of a camera user interface **1902**, as shown in FIG. **19**, as part of the reply camera function.

(165) FIG. **19** is a user interface diagram depicting the camera user interface **1902**, according to some examples, that may be presented by an interaction client **106**, in the example form of a media application, executing on the mobile device **114**. The camera user interface **1902** may present a real-time feed from the camera of the mobile device **114**, which the user can then capture as a still digital image or a digital video by selecting an appropriate button within an interaction zone **1806** of the camera user interface **1902**.

(166) The camera user interface **1902** displays a second user input content item **1904**, such as, for example, a digital image or digital video captured by the camera of the mobile device **114**. The second user input content item **1904** may comprise image data, video data, audio data, or a combination of these data types, together with augmentation or modification data (e.g., stickers, filters or lenses).

(167) The camera user interface **1902** also includes an interaction zone **1806** with a function or tools section, in the example form of a function carousel **1906**. The function carousel **1906** presents various tools and functions that may be invoked by a user of the mobile device **114** to interact with other users of the interaction system **100**, to interact with content made accessible via the interaction system **100**, and also to modify a captured second user input content item **1904** displayed within the camera user interface **1902**. The interaction zone **1806** includes a content augmentation selection graphical element **1908** (e.g., a user-selectable icon) that is user-selectable to invoke an augmentation function from a collection of augmentation functions. A selected augmentation function may then operate to augment or modify the second user input content item **1904**.

(168) FIG. **20** is a flowchart illustrating a method **2000**, according to some examples, to enable an active, viewing user of the interaction system **100** to apply a content augmentation, previously applied to a third-party user input content item by a third-party user, to a content item of the active user.

(169) The method **2000** commences at opening loop block **2002** and then progresses to block **2004** with a media application, in the example form of the interaction client **106** of the interaction system **100**, causing presentation of a third-party user input content item **1804** within the viewing user interface **1802** of the mobile device **114**. The third-party user input content item **1804** may be part of a set or collection of third-party user input content items that are published by a third-party user of the interaction system **100**. In such examples, an active user of the interaction client **106** is enabled, within the context of the viewing user interface **1802**, to scroll in both backward and forward directions through the collection of third-party user input content items, or to automatically or manually advance the display of the back of third-party user input content items.

(170) At block **2006**, the augmentation system **208** of the interaction system **100** detects or determines a first content augmentation of the third-party user input content item **1804**, the first content augmentation being the output of a first content augmentation function of the interaction system **100**. As described above, the augmentation system **208** may provide a number of augmentation or modification functions that a user can invoke to augment a digital content item. The particular first content augmentation (e.g., a particular filter or lens) that is applied to the third-party user input content item **1804** (e.g., a digital image published on the interaction system **100** by the third party) may be identifiable from metadata associated with the third-party user input content

item, and that is accessed and read by the augmentation system **208** at block **2006**. The content augmentation function detected may, for example, be an image lens function or an image filter function.

(171) At block **2008**, and responsive to determining or detecting the first content augmentation at block **2006**, the interaction client **106** automatically (e.g., without specific user invocation, or without user action on the user interface of the media application) causes presentation or display, on the mobile device **114**, of the augmentation camera graphical element **1808**. The augmentation camera graphical element **1808** is user-selectable to invoke a reply camera function of the interaction client **106** and further to cause presentation of the content augmentation selection graphical element **1908** that is user-selectable to invoke an associated first content augmentation function (e.g., the specific augmentation that was detected as having been applied to the third content item at block **2006**).

(172) At block **2010**, the interaction client **106** detects or determines user selection of the augmentation camera graphical element **1808**, which then invokes, at block **2012**, a camera function of the interaction client **106**. At block **2014**, the camera function enables the user to select or capture a user input content item (this may be referred to as the “second user input content item,” with the third-party user input content item being the “first”), and at block **2016**, causes the presentation of a content augmentation selection graphical element **1908**. The camera function, as noted with respect to FIG. **19**, presents the camera user interface **1902** that allows the active user to capture and select the second user input content item **1904** by the selection of a capture icon within the interaction zone **1806**. Further, the content augmentation selection graphical element **1908** is presented within the function carousel **1906** of the interaction zone **1806** of the camera user interface **1902**. The content augmentation selection graphical element **1908**, in the example form of a lens icon, corresponds to the first content augmentation of the third-party user input content item, detected at block **2006**. User selection of the content augmentation selection graphical element **1908**, which is associated with a particular augmentation function (e.g., lens) of the interaction client **106** within the camera user interface **1902**, invokes the associated augmentation function, and conveniently allows the active user to apply the relevant augmentation to the second user input content item **1904** within the context of the camera user interface **1902**.

(173) Where multiple augmentations are applied to the third-party user input content item **1804**, each of these content augmentations may be detected by the augmentation system **208** of the interaction system **100**, in which case multiple content augmentation selection graphical elements **1908**, corresponding to the detected multiple augmentations, may be presented within the function carousel **1906** for user selection to conveniently and selectively apply each of these multiple augmentations to the second user input content item **1904**. The method **2000** then concludes at closing loop block **2018**.

(174) FIG. **21** is a user interface diagram, showing a viewing user interface **2104**, according to some examples. The viewing user interface **2104** shows a photograph of a cat, with an augmentation in the form of a lens applied to it. The lens overlays a first augmentation **2110** in the form of sunglasses onto the eyes of the cat and adds a second augmentation **2112** in the form of the words “DEAL WITH IT” as an overlay onto the image.

(175) The viewing user interface **2104** has an interaction zone **2106** including a camera graphical element in the example form of a camera button **2102**. The camera button **2102** invokes a lens access experience **2108** (as is diagrammatically illustrated in FIG. **21**), whereby an active user of the interaction client **106** can apply a lens (e.g., the sunglasses lens with the words “DEAL WITH IT”, as shown in FIG. **21**), as applied to the third-party user input content item (e.g., the cat photo), to another, second user input content item, such as a digital image captured by the active user using a camera function of the interaction client **106**.

(176) Accordingly, user selection of the camera button **2102** opens a modular reply camera of the interaction client **106** within which the sunglasses and associated text lens is preselected, or

presented for convenient user selection, for application to another content item.

(177) In FIG. **21**, the two augmentations **2110**, **2112** are applied as part of the same lens. However, it will be appreciated that, for example, the sunglasses augmentation and the “DEAL WITH IT” text augmentation may be two separate lenses, in which case they may be presented separately for convenient user selection (the user may wish to select only one of the lenses, for example).

(178) FIG. **22** is a user interface diagram illustrating a viewing user interface **2202**, according to some examples. FIG. **22** illustrates an example in which an augmentation **2204**, in the example form of a lens, is detected as having been applied to a third-party user input content item within the viewing user interface **2202** (in FIG. **22**, the lens was applied to overlay “dog features” on the faces and heads of the individuals shown in the image). Based on this detection, a modified camera button, in the form of lens button **2206** that identifies the detected lens, is generated and caused to be displayed within the interaction zone of the viewing user interface **2202**.

(179) In this example, the lens button **2206** is a lens thumbnail that alerts the user to the fact that selection of the lens button **2206** will activate the functionality described below. The lens thumbnail is therefore indicative of the content augmentation function that has been used by the third party.

(180) Turning now to FIG. **23**, which shows a reply camera user interface **2302**, user selection of the lens button **2206** within the viewing user interface **2202** invokes a lens access function, in terms of which a modular reply camera function is activated that causes the display of the reply camera user interface **2302**. The reply camera user interface **2302** includes, within the interaction zone, a lens carousel **2304** (augmentation carousel) populated with a scrollable set of lens icons or buttons, including a lens button **2306** corresponding to the lens detected as having been applied to the third-party user input content item (the image shown in FIG. **22**). In this example, the lens button **2306** is preselected and the associated augmentation (e.g., the augmentation **2308**) is automatically applied to a user input content item displayed within the reply camera user interface **2302**. Specifically, as shown in FIG. **22**, the augmentation **2204** is a dog filter applied to the humans in the image, and in FIG. **23**, the augmentation **2308** is also the dog filter, in that case applied to the single human shown in the camera user interface **2302**. In some examples, the content augmentation function used by the third party may thus be pre-selected in the reply camera function, thereby allowing for automatic creation of an augmented version of the second user input content item with the content augmentation detected as having been applied by the third party. This allows the user to conveniently and automatically access the same augmentation used by a third-party user of the interaction system **100**

(181) A remix button **2310** is shown in a second, adjacent position within the lens carousel **2304** and is user-selectable to invoke the remix function, as described elsewhere.

(182) FIG. **24** is a user interface diagram illustrating a viewing user interface **2402**, according to some examples. FIG. **24** illustrates an example in which an augmentation **2404**, in the example form of a lens, is detected as having been applied to a third-party user input content item within the viewing user interface **2402**. In FIG. **24**, the lens is a dog filter. Based on this detection, a modified camera button in the example form of the lens button **2206**, which corresponds to the detected lens, is generated and caused to be displayed within the interaction zone of the viewing user interface **2402**.

(183) The lens button **2206** may be a generic lens thumbnail or may be a thumbnail for the specific lens or augmentation detected as having been applied by the third party. For example, if the third-party user input content item is detected as including the dog filter, the lens button **2206** may be generated or adapted to graphically illustrate the dog filter, e.g., through a suitable thumbnail or dog image. In this way, a user may be more easily alerted to the option of applying the same augmentation as the augmentation used by the third party.

(184) User selection of the lens button **2206** invokes a lens access function, in terms of which a modular reply camera is activated and which causes display of a reply camera user interface **2502**.

(185) FIG. 25 shows the reply camera user interface **2502**, according to some examples. The reply camera user interface **2502** includes, within the interaction zone, a lens carousel **2504** that is automatically presented and is populated with a scrollable set of lens icons or buttons including a lens button **2506** and a remix button **2508**. A standard camera capture button **2510** is preselected. In this example, no lens or augmentation is preselected, and the lens button **2506** is displayed in a first position and the remix button **2508** is displayed in a second position within the lens carousel **2504**. (186) As described above, the remix button **2508** will only be displayed if the third-party user input content item of the viewing user interface **2402** is appropriately permissioned for use in creating a combined user input content item (e.g., is authorized by the third-party content creator for use in a remix function).

(187) In addition to allowing the user to conveniently use the lens or augmentation used by the third party, the modular reply camera interface may thus also include (e.g., in its augmentation or lens carousel) a combination graphical element. The combination graphical element (e.g., remix button **2508**) is user-selectable to invoke the combination function, or “remix” function, as described elsewhere.

(188) FIG. 26 is a user interface diagram illustrating a viewing user interface **2602**, according to some examples. FIG. 26 illustrates an example in which an augmentation **2604**, in the example form of a lens, is detected as having been applied to a third-party user input content item within the viewing user interface **2402**. Based on this detection, a modified camera button in the example form of the lens button **2206**, which corresponds to and identifies the detected lens, is generated and caused to be displayed within the interaction zone of the viewing user interface **2602**.

(189) While examples herein refer to augmentations such as lenses applied to digital images or videos, it should be appreciated that “augmentation” may also, in some examples, refer to an audio item added to content. For example, a viewing user may select a third-party user input content item comprising an audio item (e.g., a specific song linked to an image or video clip), and use the modified camera button to combine the audio item with, or add the audio item to, a user input content item selected or created by the user.

(190) Turning to the user interface diagram shown in FIG. 27, user selection of the lens button **2206** in FIG. 26 invokes a lens access function, in terms of which a modular reply camera is activated and which causes display of the reply camera user interface **2704**. The reply camera user interface **2704** includes, within the interaction zone, a lens carousel **2706** that is automatically presented and is populated with a scrollable set of lens icons or buttons including a lens button **2708** and a remix button **2710**. In this example, in addition to determining the augmentation **2604** applied to the third-party user input content item, third-party authorization for use of the third-party user input content item in the generation of a combined user input content item is also detected. Here, the remix button **2710** is automatically preselected and a corresponding remix function (also referred to as a combination function) is applied to the third-party user input content item and a user-selected content item to display a combined user input content item within the reply camera user interface **2704**. The user may then scroll the lens carousel **2706** and select the lens button **2708** to apply the corresponding lens to the combined user input content item displayed within the reply camera user interface **2704**.

(191) It will be noted that the reply camera user interface **2704** also includes a reply graphical element **2712** (indicated by the words “REPLY TO SALLY”) that is user-selectable to invoke a reply messaging function whereby the combined user input content item, displayed within the camera user interface **2704** may be communicated to the third-party user that generated and published the third-party user input content item displayed within the viewing user interface **2602**. Example messaging functionality is described with reference to the messaging system **212**. A set of layout graphic elements **2702**, as described above, is also displayed in FIG. 27.

(192) A user of a media application may wish to create combined and/or augmented user input content items in a preview state, prior to sharing the items with other users. However, a media

application may only permit a user to combine and/or augment certain items after sharing or publication has occurred. Alternatively, if available, such combination and/or augmentation may involve numerous manual steps or selections. One example technical problem of providing switchable, or iterative, preview and combination/augmentation states on a computer apparatus may be addressed by providing user interfaces that allow the user to combine, contrast or augment multiple user input content items, according to examples of the present disclosure. According to some examples, there are provided methods and systems to enable the combining (e.g., re-mixing) of user input content items within a series of interfaces that allows a user to combine different user input content items, generated by a single user or by multiple users of an interaction system, into a combined user input content item. For example, users may be able to augment, contrast or contextualize a first user input content item with a second user input content item, or even multiple user input content items, by switching between a preview interface and a combination interface any number of times before publishing a combined user input content item.

(193) FIG. **28** is an interface sequence diagram showing a sequence **2802** of interfaces, according to some examples, that may be presented by an interaction client **106** hosted on and executed by a user system **104**.

(194) The sequence **2802** includes a camera interface **2804** that enables a user of the user system **104** to create a user input content item in the example forms of a digital image or a digital video. Specifically, the camera interface **2804** is presented by a camera system **204** of the interaction system **100**, and presents a view and capture mechanism. The view and capture mechanism includes a view window **2806** that presents a user with a live image feed from any one of multiple cameras of the user system **104** and a capture button **2808** that is user-selectable to allow the user to capture one or more images from the image feed as a digital image or a digital video. An interaction zone **2810** presents a number of augmentations or modifications, supported by the augmentation system **208**, that can be applied in real-time to digital images presented within the view window **2806**. The camera interface **2804** also presents a number of camera controls **2812** related to the capture of a user input content item, such as a camera selection mechanism, a flash activation mechanism, etc.

(195) The preview interface **2814** is likewise presented, by the camera system **204** of the interaction system **100**, post-capture of a user input content item. In some examples, the user input content item may be selected from a media library or file storage and may be presented via the preview interface **2814** without the user having captured the user input content using a camera (or the user having captured or created the user input content item at an earlier point in time and stored the item for later use). The preview interface **2814** presents a preview of a captured or selected user input content item in a preview window **2816**. A preview interface may present a content item to the user for viewing, modification and/or augmentation prior to finalization and sharing of the content item, e.g., to a content collection of the user accessible to third parties, to a third-party user via a direct message, to a shared content collection, or to a public content collection.

(196) The interaction zone **2810** presents multiple tools (e.g., of the augmentation system **208**) to enable a user to augment or otherwise modify the user input content item to generate an augmented user input content item. The preview interface **2814** also presents a transmit button **2818** to invoke the communication system **210** of the interaction system **100** and transmit the user input content item presented within the preview window **2816** using the communication system **210** of the interaction system **100**. The preview interface **2814** also includes a remix button **2820** that is user-selectable to invoke a content combination function supported by a remix system **206**, which is a subsystem of the camera system **204**.

(197) The remix interface **2822** (also referred to as a combination interface) is also presented by the camera system **204**, and specifically the remix system **206**, to enable a user to capture a supplemental user input content item and combine the supplemental user input content item with a previously captured user input content item to generate a combined user input content item within a

combination window **2824**. The remix interface **2822** presents various layout tools **2826** for laying out multiple user input content items relative to each other and within certain layouts, as well as a number of the camera controls **2812**. An interaction zone **2810** of the combination window **2824** presents various augmentation and modification tools that may be applied to combined content within the context of the remix interface **2822**.

(198) In some examples, once a combined user input content item has been created, the user may be presented with the preview interface **2814** again, showing the combined user input content item, and enabling the user to combine the combined user input content item with a further user input content item to create a further combined user input content item. For example, instead of selecting the transmit button **2818** in the preview state, the user can select the remix button **2820** to switch back to the remix interface **2822** and further “remix” the content.

(199) Further details regarding some examples of interfaces of FIG. **28** are discussed with respect to FIG. **30**.

(200) FIG. **29** is a flowchart illustrating a method **2900**, according to some examples, to facilitate the generation of a combined user input content item. While some examples are described with reference to user input content items, disclosures are broadly applicable to other types of digital content. The method **2900**, in some examples, is performed by various components of the interaction system **100** described with reference to FIG. **1**.

(201) The method **2900** commences at opening loop block **2902**, and progresses to block **2904**, where the user system **104**, and specifically the interaction client **106** (as an example of a media application), causes the presentation of the preview interface **2814**, including a first user input content item (e.g., the first digital image **3004**). As noted with respect to FIG. **28**, the first digital image **3004** may have been captured using camera interface **2804** of the interaction client **106**, and then presented within the preview interface **2814** following an image or video capture operation performed by a user via the camera interface **2804**.

(202) The interaction client **106** provides a camera function supported by the camera system **204**, which causes the presentation of the camera interface **2804**. Within the camera interface **2804**, a digital image feed, from one or more cameras of the user system **104** (e.g., of a mobile device **114**, head-wearable apparatus **116** and/or computer device **118**) is displayed within the view window **2806**. A captured input is received responsive to user selection of the capture button **2808** within the camera interface **2804**, responsive to which the first digital image **3004** may be captured as an example of the first user input content item. The first digital image **3004** is then stored either locally or remotely so as to be accessible by the interaction client **106**.

(203) The preview interface is a user interface presented to the user of the media application prior to any transmission of the content item (shown in the preview interface) to third-party users of the media application, and prior to any sharing of that content item with third-party users. In some examples, when the user is viewing the relevant content in a preview mode, or preview state, the content has not been finalized or submitted/approved for publication and can only be viewed by the user.

(204) From block **2904**, the method **2900** progresses to block **2906**, where the interaction client **106** causes the presentation of a number of modification indicia (e.g., the creative tools **3008** shown in FIG. **30**) within the preview interface **2814**. As noted with respect to FIG. **30**, each of the creative tools **3008** is associated with a respective modification function of the interaction client **106**.

Further, the creative tools **3008** include a user input content combination indicium (also referred to as a combination graphical element) in the form of the remix button **2820** that is associated with a user input content combination function (e.g., a “remix” function) of the interaction system **100**.

(205) At block **2908**, the interaction system **100** detects user selection of the remix button **2820** and, at block **2910**, responsive to the detection of the user selection of the remix button **2820**, activates the combination function (e.g., the remix function). The combination function, as noted herein, enables a user of the interaction client **106** to access a second user input content item (e.g.,

newly created digital content data, such as the second digital image **3010** shown in FIG. **30**, or a stored digital content item selected from either local or remote storage accessible by the user system **104** and then displayed in the preview state). The combination function further enables the first user input content item (e.g., first content data in the form of the first digital image **3004**) to be combined with the second user input content item (e.g., second content data in the form of the second digital image **3010**) to create a combined user input content item (combined content data). (206) To this end, the user input content combination function may invoke a camera function provided by the camera system **204** of the interaction system **100**, and enable the user to capture the second digital image **3010** from a digital image feed presented within the context of the remix interface **2822** (combination interface). The first digital image **3004** and the supplemental digital image feed may be concurrently or simultaneously presented within the remix interface **2822**. User selection of the capture button within the remix interface **2822** then causes one or more digital images **3010** from the supplemental digital image feed to be captured as the second digital image **3010**. At block **2912**, the second digital image **3010** is displayed concurrently or simultaneously with the first digital image **3004** within the preview interface, subsequent to the capture, as the combined user input content item.

(207) As noted above with respect to FIG. **28**, a layout function may also be invoked by the combination function to enable a user to lay out the first digital image relative to the second digital image within the preview interface **2814**. The layout function may enable a user to manually define or specify the relative positions and sizes, as well as other characteristics (e.g., augmentations), of the first digital image and the second digital image. The layout function may also present a number of predefined layouts, which specify relative positions and sizes of the first digital image and the second digital image within the combined user input content item. In addition to specifying relative positions and sizes, a particular layout may also automatically apply a number of other visual or audio features to the combined user input content item. For example, where a particular layout has a theme, visual characteristics (e.g., augmentations and supplemental images), and audio characteristics (e.g., sounds or songs) may be automatically applied to the combined user input content item.

(208) Following presentation of the combined user input content item within the remix interface **2822** at block **2912**, a user may then elect to return to the preview interface **2814**, e.g., by the selection of the return arrow **3018** shown in FIG. **30**. In some examples, the user may automatically be presented with the preview interface **2814** again without having to select a return element or perform a return user action. At block **2912**, the user may, within the preview interface **2814**, decide on the addition of a further user input content item (e.g., the creation and addition of a further video clip) to the combined user input content item. Accordingly, the user may select the remix button **2820** and cause the detection of user selection of that remix button **2820** at block **2914**.

(209) At block **2916**, responsive to detection of user selection of the remix button **2820**, the combination function may again be activated to present the remix interface **2822**. In other words, the combination function may be reactivated for creating a further combined item. Within the context of the remix interface **2822**, the user may access both the combined user input content item and a third user input content item (e.g., a newly captured digital image or digital video content item, captured within the context of the remix interface **2822** using the camera function), and combine the previously combined user input content item with the third user input content item to create a further combined user input content item (further combined content data). A user may thus continually loop between the preview and remix interfaces to create a number of digital user input content items that may be combined into a single item. According to some examples, a challenge of a user not being able to create and modify user content combinations containing multiple content items, e.g., before any publication or sharing thereof, can thus be addressed. The method **2900** then terminates at closing loop block **2918**.

(210) FIG. 30 is an interface sequence diagram showing a sequence **3002** of interfaces, according to some examples, that may be presented by an interaction client **106** hosted on and executed by a user system **104**. The sequence **3002** of interfaces are some examples of the more general interfaces described with reference to FIG. 28. For the sake of simplicity, the functions of the preview interface **2814** and the remix interface **2822** are described with respect to digital images, which may comprise single or multiple still digital images or a sequence of digital images that constitute a digital video clip.

(211) The sequence **3002** presents two views of the preview interface **2814**, and a single view of a remix interface **2822**. The first view of the preview interface **2814** shows a first digital content item, in the form of digital image **3004**, that is displayed within the preview window **2816**. The digital image **3004** may be a user input content item captured and created using the interaction client **106**. In FIG. 30, the digital image comprises an original digital image of fruit captured using the camera interface **2804**, to which a “fruit” augmentation was applied, within the context of the camera interface **2804** or the preview interface **2814**, to create an augmented digital image in the form of the digital image **3004**. The fruit augmentation **3022** has the effect of overlaying a character that resembles an avocado over the original image (photograph) of the fruit.

(212) The preview interface **2814** has an interaction zone **2810** that presents a collection of augmentations that can be applied to the digital image **3004**. The interaction zone **2810** also allows a user to select additional digital content to be included in a transmission to a receiving user of the interaction system **100** (e.g., by selection of the “plus” button in the media list **3020** before selection of transmission indicia in the form of the send button **3006**).

(213) A menu of user-selectable indicia for creative tools **3008** is displayed on the right-hand side of the preview interface **2814**. These creative tools **3008** enable a user to perform various modifications to the digital image **3004** (e.g., adding text, drawing on, cropping, adding music to, and inverting). The menu also includes a user-selectable indicium in the form of a remix button **2820** that is user-selectable to invoke the remix system **206** and the remix interface **2822** (e.g., a modular remix camera) to provide the user with a combination function that allows the user to combine the digital image **3004** with a second content item, in the form of a second digital image **3010**. As shown in the second preview interface **2814** in FIG. 30, a remix tooltip **3012** is displayed in conjunction with the remix button **2820** in order to inform a user regarding the functionality invoked by the remix button **2820** and to invite a user to “remix” digital content items (e.g., user input content items) to create a combined digital content item.

(214) The remix interface **2822**, also referred to as a combination interface, enables a user to capture the second digital image **3010** and presents both the first digital image **3004** and the second digital image **3010** in a layout configuration relative to each other. The second digital image **3010** may be captured within the context of the remix interface **2822** by user selection of a capture button **2808** presented within the interaction zone **2810**. The interaction zone **2810** also presents a number of augmentations that can selectively be applied by a user to the second digital image **3010**. The remix interface **2822** also includes a number of camera controls **3014** (e.g., flip camera, flash, etc.) that can be invoked for the capture of the second digital image **3010** within the context of the remix interface **2822**.

(215) The layout of the first digital image **3004** and the second digital image **3010** may be manually determined by a user (e.g., by dragging the digital images to relative positions within the remix interface **2822**) or may be selected by a user from a number of preconfigured layouts **3016** presented by the layout tool **2826**. The preconfigured layouts **3016** may include different split-screen layouts, picture-in-picture layouts, foreground/background layouts, and others, as described elsewhere.

(216) The remix interface **2822** also includes a user-selectable return element in the example form of a return arrow **3018** that is user-selectable to return a user back to the preview interface **2814**, with combined content created within the remix interface **2822**. For example, having combined the



first digital image **3004** with the second digital image **3010** within the remix interface **2822**, a user may select the return arrow **3018**. Responsive to the selection, the interaction client **106** returns displayed to the preview interface **2814**, which then presents the combined digital image. A user may then again select the remix button and be returned to the remix interface **2822**, where the user can generate or select third digital content to be combined with the previously combined content. In this way, a user can create a combination of multiple instances of user input content into a single instance or item of user input content. The provision of various augmentations that can be applied to discrete digital content components of a combined digital content item, as well as the provision of various layout options and functions, enables a user to find extensive creative expression using both user input content that is self-generated (or self-selected) and user input content obtained from other users of the interaction system **100** (or even from other sources).

(217) FIG. **31** is a flowchart illustrating a method **3100**, according to some examples, to facilitate the generation of a further combined user input content item. The method **3100** commences at opening loop block **3102** and proceeds to block **3104**, where the user system **104**, and specifically the interaction client **106** (as an example of a media application), causes presentation of a camera interface.

(218) At block **3106**, a digital image feed is presented within the camera interface. The user then selects a camera button and the media application receives the corresponding user capture input at block **3108**.

(219) Responsive to receipt of the user capture input, according to some examples, the digital image content is captured as a third user input content item at block **3110**. The method **3100** then proceeds to block **3112**, where the third user input content item is stored by the user system **104**, to be accessible by the combination function of the media application.

(220) The user then uses the combination function of the media application to create a combined user input content item. This may be done according to examples described herein. The combined user input content item is a combination of a first user input content item and a second user input content item. For example, the combined user input content item may include a first user input content item as a background image with a second user input content item being overlaid over a portion of the first user input content item. The combined user input content item is presented to the user via a preview interface at block **3114**. The preview interface includes a user input content combination indicium in the example form of a remix button, enabling the user to create a further combined user input content item.

(221) At block **3116**, the media application detects user selection of the user input content combination indicium within the preview interface. From block **3116**, the method **3100** proceeds to block **3118**, wherein the user input content combination function is activated. This may allow the user to add (e.g., overlay) the third user input content item to the combined user input content item to create a further combined user input content item.

(222) According to some examples, the user then selects the third user input content item (e.g., from a file storage or media library) and the media application receives the selection at block **3120**. The further combined user input content item is presented within the remix interface at block **3122**. As described herein, according to some examples, the user may select a layout for the further combined user input content item or manually adjust the layout of the third user input content item relative to the combined user input content item.

(223) The user may share the further combined user input content item with other users or, as described herein, and according to some examples, the user may combine the further combined user input content item with even further content items, e.g., to create an even further combined user input content item consisting of, for example, 4, 5, 6 or 7 different user input content items. The method concludes at closing loop block **3124**.

(224) A user may wish to create a combined video item in a media application to share with other users. However, media application systems may be limited in terms of the ability to combine

content items and/or the manner in which items can be combined. For example, a media application may be unable to provide a user with flexible temporal relationship selections in generating a combined video item from constituent portions. One example technical problem of providing a system that allows the user to augment, contrast or contextualize a first user input content item with a second user input content item or even multiple user input content items in a time-based manner, a spatially relative manner or a combination thereof, is addressed by a method and system according to examples of the present disclosure. According to some examples, there are provided methods and systems to enable the combining (e.g., re-mixing) of user input content items within a series of interfaces that allows a user to combine different user input content items generated by a single user or by multiple users of an interaction system, into a combined user input content item in a time-based manner, a spatially relative manner or a combination thereof. In some examples, the user can generate a combined user input content video based on one or more combination options, e.g., a specified temporal relationship between the content items forming the combined user input content video

(225) FIG. **32** is an interface sequence diagram showing a sequence **3202** of interfaces, according to some examples, that may be presented by an interaction client **106** (e.g., a media application) hosted on and executed by a user system **104**.

(226) The sequence **3202** shows a number of interfaces that allow the user, for example, to capture or access a user input content video **3234** and combine the user input content video **3234**, or a portion thereof, with another content item. The interfaces include a camera interface **3212**, a view interface **3214**, and a remix interface **3216**. Each of these interfaces provides specific functionality within the interaction client **106** and enables a user to capture, view, or modify a content item (e.g., user input content video **3234**). Each of these multiple interfaces further includes a user input content combination indicium **3218**, which is user selectable to invoke a combination function that presents the trimming interface **3206** and the remix camera interface **3222** also shown in FIG. **32**. The user input content combination indicium **3218** may be in the form of a “remix” button, “add to” button, “combine” button, or the like.

(227) The camera interface **3212** enables a user of the user system **104** to create a user input content item in the example forms of a digital image or a digital video. The user may also use the media application to add audio to the item, e.g., an audio track selected to play contemporaneously with a digital video.

(228) The camera interface **3212** is presented by a camera system **204** of the interaction system **100**, and presents a view and capture mechanism. Specifically, the view and capture mechanism includes a view window that presents a user with a live image feed from any one of multiple cameras of the user system **104** and a capture button **3240** that is user-selectable to allow the user to capture one or more images from the image feed as a digital image or a digital video. A timeline within the camera interface **3212** presents a number of augmentations or modifications, supported by the augmentation system **208**, that can be applied in real-time to digital images presented within the view window of the camera interface **3212**. The camera interface **3212** also presents a number of camera controls **3228** related to the capture of a user input content item, such as a camera selection mechanism, a flash activation mechanism, etc.

(229) The view interface **3214** enables a user to view a content item (e.g., the user input content video **3234**) retrieved from memory of the user system **104**, where the content item is stored as a result of having been received in a message or publications stream from a further user of an interaction system **100**, or as a result of having been previously captured by the user system **104** and stored in the memory.

(230) The remix interface **3216** enables a user to combine multiple content items (e.g., user input content images, user input content videos, audio, augmented reality augmentations, etc.), some of which may have been received from a different user of the interaction system **100**, to create a combined user input content item (e.g., a combined user input content video clip).

(231) The remix interface **3216** is also presented by the camera system **204**, and specifically the remix system **206**, to enable a user to capture or select a supplemental user input content item and combine the supplemental user input content item with a previously captured or selected user input content item to generate a combined user input content item within a combination window. The remix interface **3216** presents various layout tools for spatially arranging (e.g., laying out or overlaying) multiple user input content items relative to each other and within certain predefined or freeform layouts, as well as a number of the camera controls. A remix interface **3216** also presents various augmentation and modification tools that may be applied to combined content within the context of the remix interface **3216**. Further details regarding some examples of the interfaces of FIG. **32** are discussed with respect to subsequent figures.

(232) The trimming interface **3206** is likewise presented, by the camera system **204** of the interaction system **100**, post receipt, post selection, or post capture of a user input content item. The trimming interface **3206** presents a “preview” of a user input content item in a preview window **3208**. A timeline **3204** presents multiple tools (e.g., of the augmentation system **208**) to enable a user to augment or otherwise modify the user input content item to generate an augmented user input content item. The trimming interface **3206** also presents a check indicium **3210** to invoke the communication system **210** of the interaction system **100** and cause the user input content item to be presented within the remix camera interface **3222**. The trimming interface **3206** also includes a remix button that is user-selectable to invoke a content combination function supported by a remix system **206**, which is a subsystem of the camera system **204**.

(233) The remix camera interface **3222** enables a user to capture further digital content (e.g., content item **3236**) and to add, append or otherwise “mix” the content item **3236** with the user input content video **3234** in order to create the combined user input content item **3238**. The remix camera interface **3222** also includes a progress bar **3220** indicating the relative duration of each of the content items within the combined user input content item **3238** relative to each other and relative to an overall duration of the combined user input content item **3238**. To enable the capture of further digital content, the remix camera interface **3222** includes a capture button **3226** presented in a camera window **3224**. A media library selector **3230** enables a user to select and retrieve stored digital content, while an augmentation selector **3232** enables a user to select and apply any one of multiple augmentations to content portions of the combined user input content item **3238**, either individually or in aggregate.

(234) In some examples, and as described below with reference to FIG. **37** and FIG. **38**, the sequence **3202** may also include a reordering interface **3802**, enabling the user to adjust the order of two or more content items to appear in the combined user input content item.

(235) Referring again to FIG. **32**, the combined user input content item **3238** is then made available after creation within the remix camera interface **3222** for sharing (e.g., with other users of the interaction system **100**).

(236) FIG. **33** is a flowchart that describes a method **3300**, according to some examples. The method **3300** commences at block **3302** and progresses to block **3304**, where, using at least one processor of the user system **104**, the media application causes presentation of a user input content video **3234**.

(237) At block **3306**, a user-selectable user input content combination indicium **3218** is presented by the media application and, at block **3308**, the media application detects user selection of the user input content combination indicium.

(238) Selection of the user input content combination indicium **3218** activates a user input content combination function (block **3310**). The user input content combination function causes presentation of a timeline (see, for example, the timeline **3204** shown in FIG. **35**) for the user input content video **3234** within the trimming interface **3206**. The timeline **3204** permits the user to select a portion of the user input content video **3234**.

(239) At block **3312**, the method **3300** comprises detecting selection of a portion of the user input

content video **3234**. This detection may include detecting a user action with respect to the timeline **3204** for the user input content video **3234**. The causing of the presentation of the timeline **3204** may comprise causing presentation of a trim function with respect to the user input content video **3234** to enable the user to trim the user input content video to generate the portion of the user input content video **3234**.

(240) In some examples, the trim function defaults to the selection of a predetermined portion (e.g., the first five seconds) of the user input content video **3234**. (It will be appreciated that, in examples where the user input content video **3234** is of a certain duration, e.g., shorter than 5 seconds, no trim function may be presented and the user may not be enabled to select only a portion of the user input content video **3234**, with the entire user input content video **3234** being selected for combination as described below.)

(241) In some examples, the user input content combination function causes presentation of a progress indicator in the example form of a progress bar **3220** indicating a duration of the portion of the user input content video **3234** relative to a duration of a combined user input content item **3238**.

(242) At block **3314**, the user input content combination function accesses a second content item **3236**. The second content item **3236** may be a second video clip.

(243) The method **3300** may include, at block **3316**, detecting selection, by the user, of one or more combination options. The combination options may include an order selection made by the user, indicating for example whether the user input content video **3234** is to appear at the beginning or end of the combined user input content item **3238**, or whether a particular item should appear before or after the other one. The order selection may thus be indicative of a selected temporal relationship between the user input content items constituting the combined user input content item.

(244) The combination options may include a splitting selection made by the user, indicating for example, whether the user input content video **3234** is to be split into two or more separate parts (e.g., the user input content video **3234** may be split into two parts, with one part appearing at the beginning of the combined user input content item **3238** and another part appearing at the end of the combined user input content item **3238**, with a second user input content item appearing between the two parts.) Likewise, the splitting selection may indicate that the second item **3236** is to be split into two or more separate parts. The separate parts may be arranged in any order within the combined user input content video

(245) At block **3318**, the user input content combination function combines the portion of the user input content video **3234** with the second content item **3236** to create a combined user input content item **3238** (e.g., a combined user input content video). The combination may be carried out based on the one or more combination options selected by the user, e.g., to ensure that the video portions or clips are arranged in a particular order, and the method concludes at closing loop block **3320**.

(246) It will be appreciated that while examples focusing on the creation of a combined video item are described, similar techniques may be applied to create other combinations, e.g., a combined slideshow where each slide has a defined timing.

(247) Further details regarding some examples will now be described with reference to interface diagrams illustrated in FIG. **34** to FIG. **38**.

(248) FIG. **34** to FIG. **36** are interface diagrams showing a sequence of user interfaces that may be generated and displayed by the interaction client **106**, according to some examples.

(249) As shown in FIG. **34**, a view interface **3214** presents a user input content item, in the form of a user input content video **3234** depicting a dancing person. The view interface **3214** may be presented as a result of a user selecting the user input content video within a feed (e.g., a content item feed published by another user of the interaction system **100**) or having received the user input content video within a message from another user of the interaction system **100**.

(250) Within the view interface **3214**, a user may select a context menu item, or perform some other action, such as a press and hold, with respect to the user input content video **3234**, responsive to which the interaction client **106** causes the presentation of a context menu **3402** including a number of options with associated text and user-selectable indicia. The user-selectable indicia are shown to include two examples of user input content combination indicia, namely a remix button **3408** and an add button **3404**. The remix button **3408**, as described elsewhere, may enable a user to combine multiple content items into a collage or otherwise spatially arranged collection of content items that are concurrently displayed (e.g., in a split screen configuration or/and overlay configuration). The add button **3404**, according to some examples, may enable a user to combine multiple content items for sequential display (e.g., as a continuous video comprised of multiple content items, or parts thereof), with the user being enabled to specify or adjust the temporal relationship between the items.

(251) User selection of the add button **3404** from the content menu **3402** invokes the trimming interface **3206**, which is shown in FIG. **35** and which displays a full-screen preview of the user input content video **3234**, together with trimming controls. The trimming controls, in some examples, comprise a timeline **3204** having a pair of handles **3502**, and a playhead marker **3504**. User manipulation of the handles **3502** allows the user to trim the original user input content video **3234** to a desired length and, in this way, select a portion of the original user input content video **3234** for inclusion within a combined user input content item **3238**. The trimming controls may default to selecting a predetermined portion of the original user input content video **3234** (e.g., the first five seconds) or to selecting the full clip, where the original user input content video **3234** is less than a predetermined length (e.g., five seconds).

(252) A check indicium **3506**, in the example form of a checkmark, is displayed adjacent the timeline **3204**, and is user-selectable to indicate the completion of a trimming operation by the user within the trimming interface **3206**, in which case the interaction client **106** advances to the trimming interface **3206**. Should a user drag the handles **3502** to select the full duration of the original user input content video **3234**, the check indicium **3506** is disabled because the user does not have any time within a fixed duration of the combined user input content video to any additional content items (e.g., an additional video clip recorded by the user within the context of the remix camera interface **3222**.)

(253) Having completed a selection of a length (e.g., portion) of the user input content video **3234** for inclusion within the combined user input content video, or where a length is defaulted to or automatically selected by the messaging application, a user may then selectively activate the check indicium **3506** to advance display to the remix camera interface **3222**, as shown according to some examples in FIG. **36**. The remix camera interface **3222** presents a modular camera screen to the user. The remix camera interface **3222** presents a continuous image feed from one or more cameras of the user system **104**. The remix camera interface **3222** presents a capture button **3606** that is user selectable to record digital content (e.g., a still image or a video clip) as a second content item **3236** from the continuous image feed. The captured content item **3236** is then appended or added to, or otherwise “mixed with,” the selected portion of the user input content video **3234** in order to constitute the combined user input content item **3238**. The capture button **3606** may be used to capture any number of additional digital content items that may then be combined with the selected portion of the user input content video **3234** within the combined user input content item **3238**.

(254) In addition to enabling a “live” capture using the modular camera of the remix camera interface **3222**, the remix camera interface **3222** also includes a media library selector **3610** that enables a user to select additional digital content from a media library (e.g., a local library of the user system **104** or a remote server-based library of the interaction system **100**) for inclusion within the combined user input content item **3238**.

(255) The trimming interface **3206** and remix camera interface **3222** further include a progress bar **3508** that initially presents a visual display of the duration of the selected portion of the user input

content video **3234**, relative to an overall duration of a possible combined user input content item. Specifically, the highlighted portion of the progress bar visually depicts the duration of the selected portion of the user input content video **3234**. As additional content items (e.g., the content item **3236**) are added to the combined user input content item **3238** within the context of the remix camera interface **3222**, the progress bar **3508** updates to indicate an overall occupied duration of a potential maximum duration of the combined user input content item **3238**. The progress bar **3508** may display the duration of each of the content items that constitute the combined user input content item **3238** in a different color or using visual differentiation for each content item duration. (256) The remix camera interface **3222** further includes an augmentation selector **3612** which is user selectable to present a scrolling list of augmentations (e.g., filters or stickers) that a user can optionally apply to a captured content item **3236**. A check indicium **3602** is displayed within the remix camera interface **3222** and remains disabled until a user has identified at least one content item **3236**, either through capture or retrieval, for inclusion within the combined user input content item **3238**. The check indicium **3602**, once active, is user selectable to indicate completion of the creation of the combined user input content item **3238**, following which the combined user input content item **3238** is made available within a viewing interface for previewing, prior to communication to other users of the interaction system **100**, either by direct messaging or sharing on a publication feed of the user supported by the interaction system **100**.

(257) An undo button **3604** within the remix camera interface **3222** is user selectable to return a user from the remix camera interface **3222** to the trimming interface **3206**. Within the remix camera interface **3222**, camera controls **3608** enable a user to apply various controls when capturing a content item **3236**.

(258) It will also be noted that each of the view interface **3214** and the remix camera interface **3222** include attribution data **3406** that indicates and identifies a source (e.g., a user of the interaction system **100**) and context of a displayed user input content item (e.g., a user input content video **3234**). The attribution data **3406** may be displayed on playback of a combined user input content item **3238** for each of the constituent digital content items so as to provide the source and context information for a particular section or portion of the overall combined user input content item **3238**. Within any of the interfaces, the attribution data **3406** is user-selectable to present a profile page (e.g., a user profile page) for a particular source, if such a profile page exists and is published according to appropriate permissions received from the relevant source.

(259) In some examples, when the combination function is activated or invoked, it may cause simultaneous presentation of the first user input content video and a digital image feed from the user's camera, e.g., within the combination (remix) interface. The camera function, as invoked by the combination function, may receive user capture input and, responsive to receipt of the user capture input, capture digital content as the second user input content item. The combination function may cause simultaneous presentation of the first user input content video (or part thereof) and the second user input content item, e.g., in the combination interface.

(260) FIG. **37** and FIG. **38** are interface diagrams showing user interfaces that may be generated and displayed by the interaction client **106**, according to some examples.

(261) As mentioned above, one of the combination options related to the creation of a combined user input content item, e.g., a combined video, may be an order selection. The order selection, in some examples, is an instruction relating to the temporal arrangement of the parts or portions of the combined user input content video. According to some examples, a user may be permitted to reorder user input content items in the example form of video clips.

(262) FIG. **37** shows a camera interface **3702** according to some examples. The camera interface **3702** may be presented to allow the user to capture digital content using a camera button **3708**. The camera interface **3702** also includes an undo button **3706** and a preview button **3710**.

(263) Referring to the thumbnails **3704** shown in FIG. **37**, the camera interface **3702** displays the thumbnails **3704** to identify and indicate a sequence of video clips already captured or selected by

the user. These clips may, for example, have been captured and/or selected in the process of creating a combined user input content video item. In FIG. 37, the thumbnails 3704 indicate that the user has captured or selected a sequence of four clips: a first clip of 2.1 seconds, a second clip of 1.2 seconds, a third clip of 4.3 seconds, and a fourth clip of 2.2 seconds. The clips are shown in a particular order, as they would appear in the finalized, combined user input content video, should the user choose to publish it in its current form.

(264) FIG. 37 also illustrates a progress bar 3712 according to some examples, shown in the camera interface 3702. The user is able to play the clips in their current sequence in the camera interface 3702, and the progress bar 3712 provides a visual indication of the playback position, as well as the duration of the portions of each clip relative to the combined video.

(265) In response to detecting a reordering selection made by the user, a reordering interface 3802 may be generated and presented. The reordering interface presents the content items making up the combined user input content video, e.g., the constituent clips, in their current order, and enables the user to modify the temporal relationship between the items in the sequence. The reordering selection may, for example, be made by pressing and holding on one of the clips forming part of the thumbnails 3704, performing a dragging gesture in respect of one of the clips, or selecting a dedicated clip order button or option (not shown). Responsive to the reordering selection being made, the media application may thus activate a reorder mode or reorder state, presenting the reordering interface 3802.

(266) FIG. 38 shows the reordering interface 3802, according to some examples. In the reorder state, the user is permitted to reorder video clips, delete video clips, watch a playback of the clips, and edit or augment one or more of the video clips.

(267) In some examples, the media application may initiate haptic feedback to notify the user that the reorder mode or state has been activated. The media application may cause the clips 3804 to move relative to the rest of the elements in the reordering interface 3802 (e.g., through a “wiggling” motion) to notify the user that the reorder mode or state has been activated.

(268) The adjustable sequence of clips 3804 can be reordered by dragging a clip into the desired position in the sequence of clips 3804. The user can delete a clip by tapping the “(-)” icon displayed in the top right corner of each thumbnail. When a user is done, the check mark button, or tick button 3806, is selected.

(269) It will be appreciated that this reordering technique may be employed, in some examples, as part of the combination function of the media application to allow the user to reorder content items forming part of, or intended to form part of, a combined user input content item.

(270) Various entry points may be provided by the interaction client 106 in order to invoke the combination function. For example, the view interface 3214 shown in FIG. 34 provides two entry points: the remix button 3408 and the add button 3404. FIG. 39 and FIG. 40 show further examples of entry points. The view interface 3904 shown in FIG. 39 displays an example in which a remix tray 3902 is presented responsive to the user selecting an appropriate menu icon or performing a user activity (e.g., a press and hold operation) with respect to a displayed user input content video.

(271) The remix tray 3902 includes two user-selectable combination indicia: both an add before or after button 3906 and an add together button 3908. The add before or after button 3906 may, for example, launch the combination process described with reference to FIG. 33. The add together button 3908 may, for example, launch the combination process described with reference to FIG. 29. These buttons provide a user with options of relative temporal or visual placement of multiple content items within a combined user input content item.

(272) The camera interface 4004 shown in FIG. 40, of which the remix camera interface 3222 is an example, includes a menu presenting several layout tools 4002, which menu includes a user input content combination indicium 4006 that is user selectable to invoke the user input content combination function to allow the temporal arrangement or placement (e.g., time-based sequencing) of user input content items into a video or slideshow, as described herein. The layout

tools **4002** also include a number of spatial layout options that enable the side-by-side or overlaid layout of multiple digital media content items. The user input content combination indicium **4006** is user selectable in order to invoke the combination function and display the trimming interface **3206** and the remix camera interface **3222**.

(273) The media application may provide access to a media platform (e.g. interaction system **100** or part thereof). The user may thus be a media platform user, with the media platform facilitating sharing of the combined user input content video with a further user of the media platform. Sharing permissions related to sharing of content via the media platform may be stored for the user and the further user by the media platform, and the sharing of the user input content video by the user with the further user may be performed in accordance with the sharing permissions.

(274) A media application (e.g., a messaging application) may allow a user to view a content item and share that content item to other user devices. A content item may be shareable to a plurality of content sharing destinations. However, in some examples, the media application may be designed to restrict or prevent a user from sharing a third-party content item to one or more content sharing destinations, e.g., a user may be discouraged from publishing a third-party image directly to the user's own content collection by disabling or deactivating the relevant content sharing destination when the user is viewing that third-party image. While restrictions of this nature may encourage users to be creative and share original content, these restrictions may result in user dissatisfaction and even reduced usage of the media application. One example technical problem of unlocking previously locked or deactivated content sharing destinations based on user interaction with the media application may be addressed by providing a method and system with improved functionality, automatically linking the creation of combined user input content items to presentation of and interaction with content sharing destinations.

(275) FIG. **41** is an interface sequence diagram showing a sequence **4102** of interfaces, according to some examples, that may be presented by an interaction client **106** hosted on and executed by a user system **104**.

(276) The sequence **4102** includes a viewing interface **4104** that enables a user of the user system **104** to view user input content items published and/or shared by third parties. The third-party user input content items may, for example, be content forming part of content collections, e.g., a third-party user input content item may form part of a “personal story” of a third-party user that has been created and sent/broadcast by that user, or may form part of a public feed of curated third-party items presented to the user by a media application.

(277) A view window **4106** presents a user with a view of these one or more user input content items, e.g., as a sequential set of content items presented by the media application on the user system **104**. An interaction zone **4110** presents a number of options available to the user in the view window **4106**. For example, the interaction zone **4110** may include a user-selectable button allowing the user to react to the content (e.g., “like” the content) and another user-selectable button allowing the user to comment on or initiate a message to the third-party user associated with the content. As shown in FIG. **41**, according to some examples, the interaction zone **4110** includes a user-selectable content sharing graphical element **4108**. The content sharing graphical element **4108** may, for example, be implemented as a “Share” button or a “Send to” button.

(278) Selection of the content sharing graphical element **4108** causes presentation of a sharing interface **4112**. Such an interface may, for example, be implemented as a “Send To” interface or “Send To” tray presenting the user with selectable options related to sharing of the content item viewed by the user.

(279) The sharing interface **4112** may be presented by the communication system **210**, to enable a user to share a content item presented in the viewing interface **4104** with other users. To this end, the sharing interface **4112** presents various sharing options, referred to herein as “destinations,” identified by respective destination graphical elements **4114**, **4116**, **4118a** in the sharing interface **4112**. The destination graphical elements **4114** and **4116** are user-selectable to enable the user to



share the relevant content item to the sharing destinations associated with those destination graphical elements **4114** and **4116**. For example, the destination graphical element **4114** may identify a first sharing destination such as “share this image with a friend in a private message,” while the destination graphical element **4116** may identify a second sharing destination such as “share this image via email.” If the relevant destination graphical element **4114**, **4116** is selected by the user, the sharing of the content item to the destination associated with the particular destination graphical element **4114**, **4116** is initiated.

(280) A content item being viewed by a user may, in some examples, not be shareable to one or more content sharing destinations. For example, the interaction system **100** may be configured such that a user is not permitted to share a third-party user input content item directly to the user's “story” or directly to a public feed/collection. Such a restriction may, for example, be implemented to prevent users from simply re-sharing or re-publishing content already shared or published by other users of the interaction system (or make it more difficult for them to do so). For example, the interaction system may be designed to facilitate the sharing or publication of “original” or “first-time” content items to stories and/or public collections/feeds, while preventing or reducing the sharing or publication of “non-original” or “existing” content items back for presentation to other users of the interaction system. To this end, the sharing interface **4112** may present a destination graphical element **4118a** identifying a locked destination. For example, the **4118a** may alert the user that a certain content sharing destination, such as “sharing to your story,” is not available (locked or inactive) for the particular content item the user is viewing.

(281) Various content sharing destinations may be available or presented to a user in examples of the present disclosure. These may include, for example, in the case of a user using the interaction client **106**, sharing via direct message to another user of the interaction client **106**, sharing to the user's story or other content collection, sharing to a public content collection available to users of interaction clients **106**, sharing to third-party applications such as third-party messaging applications and/or third-party social networking platforms, sharing via email, sharing via text or multimedia message, and the like.

(282) As shown in FIG. **41**, the destination graphical element **4118a** is displayed in conjunction with a remix button **4120**. In some examples, the remix button **4120** allows the user to unlock or activate the locked destination by re-mixing the relevant content item, thereby creating a new, combined content item that is shareable to the previously locked or unavailable destination.

(283) A remix interface **4122** (also referred to as a combination interface) is presented by the camera system **204** and specifically the remix system **206**, responsive to user-selection of the remix button **4120**, to enable a user to capture a further content item and combine the further content item with the third-party content item viewed in the viewing interface **4104**, to generate a combined user input content item within a combination window **4126**. The remix interface **4122** presents various layout tools **4128** for laying out multiple content items relative to each other and within certain layouts, as well as a number of camera controls **4124**. An interaction zone **4110** of the combination window **4126** presents various augmentation and modification tools that may be applied to combined content within the context of the remix interface **4122**.

(284) In some examples, once a combined content item has been created, the user may be presented with the sharing interface **4112** again, presenting the various sharing destinations, but with the previously locked sharing destination being unlocked, represented by an updated destination graphical element **4118b**. The updated destination graphical element **4118b** is user-selectable, like the other destination graphical elements **4114**, **4116**, to enable the user to share the combined content item, created using the remix interface **4122**, to the previously unavailable content sharing destination.

(285) To enable the user to share the combined content item to the unlocked content sharing destination, the media application may thus cause presentation of an updated sharing interface **4112** with an updated destination graphical element **4118b**, and determine that the user has selected the

updated destination graphical element **4118b**. Responsive to determining the user selection of the updated destination graphical element, the media application causes the combined content item to be published to the unlocked content sharing destination.

(286) Further details regarding some examples of interfaces of FIG. **41** are discussed with respect to examples below.

(287) FIG. **42** is a flowchart illustrating a method **4200**, according to some examples, to facilitate the generation of a combined user input content item and unlocking a locked content sharing destination. While some examples are described with reference to user input content items, disclosures are broadly applicable to other types of digital content. The method **4200**, in some examples, is performed by various components of the interaction system **100** described with reference to FIG. **1**.

(288) The method **4200** commences at opening loop block **4202**, and progresses to block **4204**, where the user system **104**, and specifically the interaction client **106** (as an example of a media application), causes the presentation of a third-party user input content item. The third-party user input content item may, for example, be an image or a video clip being viewed by an active user on the user system **104**. The third-party user input content item may include multiple types of content, such as audio content combined with image content, and may include one or more augmentations or modifications, as described elsewhere herein.

(289) The third-party user input content item may be presented within a viewing interface **4104** in conjunction with a content sharing graphical element **4108**. The content sharing graphical element **4108** is user-selectable to invoke a content sharing function of the interaction client **106**.

(290) According to some examples, the method **4200** includes determining third-party authorization for the third-party user input content item to be combined with other content items to create combined user input content items at block **4206**. For example, an authorization setting related to the third-party user input content item, indicating this authorization, may be detected at block **4206**. The determining third-party authorization for the third-party user input content item to be combined may thus comprise determining a user setting related to the third-party user input content item indicating the third-party authorization. A third-party content creator, when creating or publishing the third-party user input content item, is presented with the option of either allowing or disallowing the use of the third-party user input content item in combination with other content items, created or originated with other users of the interaction system **100**. This allowance/disallowance is then recorded and stored in the database **128** as the authorization setting related to the relevant third-party user input content item, and is accessed at block **4206** to perform the detection of the third party authorization.

(291) The method **4200** proceeds to block **4208**, where user selection of the content sharing graphical element **4108** is determined. At block **4210**, responsive to the determining of the user selection of the content sharing graphical element **4108**, a content sharing function of the interaction client **106** is invoked. This causes presentation of destination graphical elements **4114**, **4116**, **4118a** on the user system **104**, for example, to display to the user various destinations to which the content the user is viewing can be sent.

(292) As mentioned with reference to FIG. **41**, each destination graphical element identifies a content sharing destination (or at least one content sharing destination). The destination graphical element **4118a** may identify a first, locked content sharing destination while the two additional content sharing destinations (identified by **4114** and **4116**) may be unlocked (available) content sharing destinations.

(293) The method **4200** may include, at block **4208**, determining or checking a content type associated with the third-party user input content item being viewed by the user. Based on the content type, the interaction client **106** may determine the content sharing destinations to present to the user and/or the availability of each content sharing destination, e.g., whether it is locked or unlocked. For example, the interaction client **106** may determine that the user has selected the

content sharing graphical element **4108** in respect of a third-party user input content item and check a sharing setting associated with the third-party user input content item in the database **128**. The sharing setting may indicate that the third-party user input content item may not be shared directly to the first content sharing destination. In some examples, the sharing setting may be specific to the user. Responsive thereto, the first content sharing destination may be presented as locked or unavailable in the sharing interface **4112**.

(294) According to some examples, the first content sharing destination is locked in respect of the third-party user input content item, or even in respect of all third-party user input content items, while additional content sharing destinations are unlocked for the third-party user input content item or all third-party user input content items. To this end, at block **4210**, the destination graphical element **4114**, **4116** identifying each respective available (unlocked) content sharing destination is presented and is user-selectable to enable the user to share the third-party user input content to the content sharing destination associated with each respective element **4114**, **4116**.

(295) According to some examples, the method includes causing presentation, by the interaction client **106**, of a combination graphical element in the form of a remix button **4120** at block **4212**. The remix button **4120** is user-selectable to invoke a combination function (or “remix” function) of the interaction client **106**, and user selection thereof is determined at block **4214**. Referring to block **4206** in FIG. **42**, it will be appreciated that, in the event that it is determined that the third party has not authorized combination of the third-party item with other items, according to some examples, no remix button **4120** is displayed.

(296) Responsive to determining the user selection of the remix button **4120**, the interaction client **106** activates the combination function at block **4216**. The combination function allows the user to access (e.g., select, capture, or create) a second user input content item and combine the second user input content item with the third-party user input content item to create a combined user input content item (block **4218**). The combination function may provide a number of sub-functions or tools to facilitate the combining of the third-party user input content item and the other user input content item. These tools may include a layout function to enable the user to specify a layout relationship of the third-party user input content item, relative to the other user input content item, or within the context of a broader combined display, during the presentation of the combined user input content item. The combination function also makes a number of the augmentations of the augmentation system **208** discussed herein available for application and use in the creation of the combined content element.

(297) In some examples, the combination function, in addition to enabling the combining of multiple content items, also enables the unlocking of the previously locked content sharing destination, at block **4220**. In some examples, once the user has created the combined user input content item (or once the user has opted to create it), the interaction client **106** causes the locked content sharing destination to become unlocked. In other words, while the user was not able to share the original third-party user input content item to that destination, the user is permitted by the interaction system **100** to share a combined user input content item, which includes the original third-party user input content item as a constituent part, to that destination.

(298) According to some examples, at block **4222**, the method **4200** includes enabling the user to share, using the interaction client **106**, the combined user input content item to the unlocked first content sharing destination, by presenting an updated destination graphical element **4118b** in the sharing interface **4112**, and causing the updated destination graphical element **4118b** to be user-selectable to initiate sharing to the (unlocked) content sharing destination associated with the element **4118b**. Alternatively, in some examples, once the combined user input content item has been created/selected, it may be automatically shared to the newly unlocked content sharing destination based on the user's initial selection of the remix button **4120**. The remix button **4120** may be positioned inside of, adjacent to, or otherwise in association with the destination graphical element of the content sharing destination to which the user wishes to share.

(299) In other words, the remix button **4120** may be positioned closer to the locked destination's destination graphical element than to the destination graphical element of the other destinations, thus alerting the user to the fact that the locked destination can be unlocked through the re-mixing operation and/or that re-mixing will automatically cause transmission to that destination. In some examples, the sharing interface may include a tooltip or other message indicating to the user that a particular content sharing destination is not available for the content item being viewed, but can be made available by re-mixing the content item. The method then concludes at closing loop block **4224**.

(300) FIG. **43** is a flowchart illustrating a method **4300**, according to some examples, to facilitate the generation of a combined user input content item and the sharing of the combined user input content item to a content sharing destination. The method **4300**, in some examples, is performed by various components of the interaction system **100** described with reference to FIG. **1**, and specifically by the interaction client **106** (as an example of a media application).

(301) The method **4300** commences at opening loop block **4302** and progresses to block **4304**, where, responsive to the user selecting the remix button **4120** (as described with reference to FIG. **41** and FIG. **42**), the remix interface **4122** is presented, and the interaction client **106** causes presentation of a digital image feed from at least one camera of the user system **104**.

(302) The third-party user input content item and a second (supplemental) digital image feed may be concurrently or simultaneously presented within the remix interface **4122**. The interaction client **106** receives user capture input at block **4306** and, responsive thereto, causes one or more digital images from the digital image feed to be captured as the second user input content item at block **4308**.

(303) According to some examples, the method **4300** includes storing the second user input content item to be accessible by the interaction client **106** at block **4310**.

(304) The method **4300** proceeds to block **4312**, where the third-party user input content item and the second user input content item are combined in order generate the combined user input content item. According to some examples, at block **4314**, the interaction client **106** causes an updated destination graphical element to be presented within the sharing interface, as described with reference to FIG. **41** and FIG. **42**.

(305) The method **4300** may include, at block **4314**, determining or checking a content type associated with the combined user input content item. Based on the content type, the interaction client **106** may determine the content sharing destinations to present to the user and/or the availability of each content sharing destination, e.g., whether it is locked or unlocked. For example, the interaction client **106** may determine that the user has created a combined user input content item, which may be classified as an “original” content item, and check the sharing setting related to items of this type in the database **128**. The sharing setting related to items of this type may, for example, include that the “original” content item may be shared with another user via a direct message, but not shared or re-published to a public feed in its original form. The sharing setting may thus provide information as to sharing permissions and/or sharing restrictions.

(306) At block **4316**, user selection of the updated destination graphical element (e.g., the element **4118b**) is detected. Responsive to detecting the user selection of the updated destination graphical element, the interaction system **100** causes the combined user input content item to be published to the unlocked first content sharing destination at block **4318**. The method concludes at closing loop block **4320**.

(307) FIG. **44** to FIG. **47** are interface diagrams showing a sequence of user interfaces that may be generated and displayed by the interaction client **106**, according to some examples.

(308) FIG. **44** is a user interface diagram showing a viewing interface **4402**, according to some examples. In FIG. **44**, an active user is viewing a third-party user input content item **4404**. The third-party user input content item **4404** is an augmented image. Specifically, the third-party user input content item **4404** consists of a photograph of a person's hand holding an ice cream cone with

a character, also holding an ice cream cone, overlaid onto a bottom right region of the photograph. The third-party user input content item **4404** forms part of a third party's content collection, referred to as a "story." The story may, for example, be presented to the active user as part of a personal feed or public feed, or in response to the user visiting the third party's profile and selecting the story. The third party's name is included in the viewing interface **4402**, as indicated by reference numeral **4406**.

(309) As indicated by the progress bar **4416**, the presentation of the third-party user input content item **4404** is part of a sequential presentation of a set of third-party user input content items.

(310) The viewing interface **4402** includes an interaction zone **4410**, presenting the active user with options for interacting with the third-party user input content item **4404** or with the third party **4406**. The user may select a camera button **4408** to launch a camera of the user system **104** of the user, e.g., to capture an image in response to the third-party user input content item **4404**. The user may also opt to send a message to the third party by selecting a reply graphical element **4412**.

(311) The interaction zone **4410** further includes a content sharing graphical element in the example form of a send button **4414**. In some examples, if the user selects the send button **4414**, and responsive to determining a combination authorization with respect to the third-party user input content item **4404**, a content sharing function is invoked. The content sharing function causes presentation of a sharing interface **4502** on the user system **104**.

(312) FIG. **45** is a user interface diagram showing the sharing interface **4502**, according to some examples. The sharing interface **4502** includes a plurality of destination graphical elements. Each destination graphical element identifies a content sharing destination that may be available to the active user. Examples of these elements are identified by reference numerals **4504**, **4506**, **4508**, **4510** and **4512** in FIG. **45**.

(313) The destination graphical element **4504** identifies a first content sharing destination. In FIG. **45**, the first content sharing destination is "Stories," meaning that selection of this option would allow the user to share the third-party user input content item **4404** to the user's story. However, in some examples, and as shown in FIG. **45**, the "Stories" destination is locked. In other words, the user is restricted from selecting the first content sharing destination (directly). Within the destination graphical element **4504**, a remix button **4514** is provided in the form of a selectable cell. A tooltip **4516** below the remix button **4514** informs the user that the destination is unavailable/locked and how to make the destination available: "Add your take to share to Stories".

(314) In other words, the user is advised, via the sharing interface **4502**, that the remix operation may be carried out to create a combined user input content item (including the third-party user input content item **4404** as a constituent part) before the content can be shared or published to the first content sharing destination (e.g., to the user's story, which is typically a content collection created and published by the user via the interaction client **106** to be accessible by a plurality of other users of the interaction client **106**). According to examples of the present invention, various different tooltips and/or messages indicating that the content sharing destination is ununlockable via the combination function, or similar indications, may be employed.

(315) A destination graphical element such as the destination graphical element **4504** may include a number of related destinations, e.g., the "Stories" destination may be a category containing a plurality of stories, e.g., the active user's personal story, a shared story and/or a public story. In FIG. **45**, none of these "sub-destinations" are available as the entire "Stories" content sharing destination is locked.

(316) The destination graphical elements **4506**, **4508**, **4510** and **4512** identify additional content sharing destinations, each of which is unlocked/available. The destination graphical element **4506** is user-selectable to enable the user to share the third-party user input content items **4404** via the MESSAGES™ application, and the destination graphical element **4508** is user-selectable to enable the user to share the third-party user input content item **4404** via the WHATSAPP™ application (both being external, or third-party, applications). The destination graphical element **4510** and the

destination graphical element **4512** are user-selectable to enable the user to share the third-party user input content item **4404** by way of direct message to another user of the interaction system **100**, via the interaction clients **106**. When selecting the destination graphical elements **4506**, **4508**, **4510** and **4512**, the third-party user input content item **4404** is shareable to the respective content sharing destinations in its original form, because these content sharing destinations are unlocked/available, in contrast to the destination graphical element **4504** which identifies a locked/unavailable destination (or multiple locked destinations) for the third-party user input content item **4404** in its original form.

(317) The user may also search for other content sharing destination using a search bar **4518** located at a top region of the sharing interface **4502**.

(318) As mentioned above, the interaction system **100** may detect a sharing setting associated with the third-party user input content item **4404**, e.g., indicating that the third-party user input content item **4404** is not directly shareable to the first content sharing destination (“Stories” in FIG. **45**). The sharing setting may indicate that the third-party user input content item **4404** is, or can become, shareable to the first content sharing destination as part of combined user input content items. In response to detecting this setting, the interaction client **106** may generate the sharing interface **4502** as described above in relation to the destination graphical element **4504**.

(319) Referring to the user interface diagram in FIG. **46**, a remix interface **4602**, also referred to as a combination interface, is presented. The remix interface **4602** is automatically activated in response to the user selecting the remix button **4514**.

(320) A remix lens, or remix augmentation, is automatically applied to merge or combine the third-party user input content item **4404** with a second user input content item **4604**, which in FIG. **46** is an image captured as a “selfie” by the active user, to create a combined user input content item **4612** within the remix interface **4602**. Accordingly, in some examples, invoking the combination function allows the user convenient access to the remix feature to create the combined user input content item **4612**, and may automatically cause presentation of a digital image feed from a camera of the user system **104**, e.g., “selfie” camera, to facilitate capturing of a second user input content item.

(321) The remix interface **4602** includes a set of layout graphic elements **4608**, each of which is user-selectable to apply a different layout to the combination of third-party user input content item **4404** with the second user input content item **4604**. Each of the layout options associated with a layout graphic element **4608** may allocate a certain section of an interface canvas to each of the images or may define a layout of the images relative to each other. The same layout options may be applied to video content. Once the user is satisfied with the second user input content item **4604** and layout, a remix capture button **4606** can be selected to finalize the combined user input content item **4612**. In FIG. **46**, the user takes a “selfie” which is overlaid onto a portion of the original third-party user input content item **4404**. However, this is merely an example of a layout used to position the two content items relative to each other.

(322) It will be noted that the remix interface **4602** also includes a contextual message **4610** that provides contextual information regarding the combination function (e.g., in this example, indicating that the active user is creating the combined user input content item **4612** based on an original content item by the user “John”). The contextual message **4610** may also be user-selectable to invoke the messaging system **212** of the interaction system and conveniently enable the active user to transmit (e.g., by way of a message or publication to a feed) the combined user input content item **4612** to the third party user (e.g., “John”) by way of a “reply” to the third party's publication of the third-party user input content item **4404**.

(323) FIG. **47** is a user interface diagram showing the sharing interface **4502**, according to some examples. Once the combined user input content item **4612** has been captured through selection of the remix capture button **4606**, the user is taken back to the sharing interface **4502**.

(324) The destination graphical elements **4506**, **4508**, **4510**, and **4512** remain the same as shown in

and described with reference to FIG. 45 (in other words, they remain unlocked and available to share to), but the destination graphical element **4504** of FIG. 45 is replaced with an updated destination graphical element **4702**. The updated destination graphical element **4702** is generated and presented by the interaction client **106**, and identifies the previously locked content sharing destination (“Stories”), which was unlocked through use of the remix function. The updated destination graphical element **4702** bears the label “SHARE TO YOUR STORY”. This alerts the user to the fact that the combined user input content item **4612** can be shared to the “Stories” destination, which was not available in respect of the sharing of the original third-party user input content item **4404**. User selection of the updated destination graphical element **4702** then results in publication of the combined user input content item **4612**, e.g., to the user's story to be accessible to a restricted set of other users of the interaction system or in some examples to a public feed.

(325) As mentioned above, a media application (e.g., a messaging application) may allow a user to view a content item and share that content item to other user devices. A content item may be shareable to different content collections (e.g., “stories” or “public feeds”), managed by the collection management system **226** of the interaction system **100** according to some examples. A user may wish to interact with a content item forming part of a collection (e.g., create a “remix” thereof) and/or may wish to view interactions of other users relating to that content item. One example technical problem of configuring a media application both to enable users to create combined content items from a base content item and enabling such combined content items to be accessed automatically, and in a consolidated manner, may be addressed by providing a method and system with improved functionality, such as those described according to examples with reference to FIG. 48 to FIG. 59.

(326) FIG. 48 is an interaction diagram illustrating a process **4800** involving devices and components of the interaction system **100**, according to some examples. While some examples are described with reference to user input content items, disclosures are broadly applicable to other types of digital content. FIG. 48 shows four user systems **104** in the example forms of a first user device **4802**, a second user device **4804**, a third user device **4806** and a fourth user device **4808**. Each device **4802**, **4804**, **4806**, **4808** has an instance of a media application in the example form of the interaction client **106** installed thereon. The user devices **4802**, **4804**, **4806**, **4808** communicate with each other and with the interaction server system **102** of the interaction system **100** (and other components of the interaction system **100** as required), using the media application.

(327) While four user devices are shown in FIG. 48, it will be appreciated that examples may involve significantly larger numbers of user devices sharing data and content with each other and to content collections via components of the interaction system **100**, using various components of the interaction system **100**, such as the collection management system **226**.

(328) Turning now more specifically to the process **4800**, at block **4810**, a first user of the media application uses the first user device **4802** to create and share a first user input content item. In examples as described with reference to FIG. 48 to FIG. 59, the first user input content item is also referred to as a “base item.” The term “base item” refers to the use of the content item to create a combined content item. The base item may, for example, be a digital image or video clip, or content data including a combination thereof, and may include any number of modifications or augmentations, as described elsewhere. The base item may also include audio content, e.g., the base item may be a relatively short video clip (e.g., 15 or 20 seconds) with an audio track, as selected by the first user, playing at the same time as the video clip.

(329) In examples according to the present disclosure, when sharing the base item, the first user specifically elects to share the base item to a first content collection, in the example form of a public or semi-public feed provided by the interaction system **100**. The first content collection may thus be a public content collection. It will, however, be appreciated that techniques described herein may be applied to processes relating various content collections.

(330) The media application may have a dedicated tab, or category, accessible by all users to view

and navigate the public feed (it will, however, be appreciated that this may also apply to private or semi-public feeds). The public feed may include a large number of content items, all being designated as “public”, e.g., accessible by all users of the interaction system **100**. In some examples, the public feed may include mainly or exclusively content in video format having a limited duration (e.g., only clips shorter than 60 seconds or 30 seconds per clip may be posted to the public feed). The interaction system **100** (e.g., using the collection management system **226**) may dynamically update the public feed, and each user of the interaction system **100** may be provided with a personalized or curated public feed, e.g., based on their activity on the media application, items they liked or interacted with, search history, relationships with other users, or combinations thereof.

(331) The first user may, for example, select a “share to public feed” button after creating the base item. The first user device **4802** causes the base item to be transmitted to the interaction server system **102** along with an indication that the base item is to be posted to the public feed (see arrow **4812** in FIG. **48**). The interaction server system **102** then makes the base item available to users of the interaction system **100** via the public feed at block **4814**.

(332) The base item then becomes available for other users of the interaction system **100** to view and interact with (e.g., “like”, share or react). If the base item is designated as “public”, in some examples, a user of the interaction system **100** may access it in various ways. For example, the user may find the base item in the user's curated public feed. Alternatively, the user may view a “live story” which includes the base item, or access the first user's profile directly to view the base item.

(333) At block **4816**, a second user of the interaction system **100**, using the second user device **4804**, accesses the base item and uses the base item to create a combined user input content item (“remixed item”). In other words, the second user creates a combined user input content item which includes the base item and one or more other content items. The second user elects to share the combined user input content item back into the public feed. The second user device **4804** transmits the combined user input content item together with a sharing instruction to the interaction server system **102** (block **4818**). Similarly, at block **4820**, a fourth user, using the fourth user device **4808**, accesses the same base item and creates another combined user input content item which is shared back into the public feed. The fourth user device **4808** transmits the combined user input content item together with a sharing instruction to the interaction server system **102** (block **4822**).

(334) The interaction system **100**, e.g., using the collection management system **226**, then makes the remixed items created by both the second user and the fourth user available in the public feed. Further, according to examples of the present disclosure and as shown in FIG. **48**, at block **4824**, the interaction system **100** makes the two remixed items available in a second content collection. The second content collection is a collection of all remixed items created from the base item. In other words, the second content collection is a set of content items, all of which are combined items generated by combining the base item with another content item. The collection management system **226** may dynamically update the second content collection in response to user of the interaction system **100** creating public “remixes” of the base item. The second content collection, like the first content collection, may be a public content collection accessible by the entire user base of the interaction system **100**. It will, however, be appreciated that techniques described herein may be applied to processes relating various content collections, e.g., semi-public content collections that have certain restrictions in terms of users allowed to access them.

(335) A third user of the interaction system **100**, using the third user device **4806**, then requests to view all “remixes” made from the base item across the user base of the interaction system **100**. For example, the third user may be viewing the base item in the public feed and be interested in seeing how other users have interacted with, reacted to, or built on, the content in the base item. The interaction system **100** may provide, on a viewing interface of the third user device **4806**, an indication to the third user that the base item has been remixed by other users (in the case of FIG. **48**, the second and fourth users) together with a user-selectable element. In response to the third



user requesting to view all “remixes,” e.g., by selecting the user-selectable element, at block **4826**, a collection interface may be presented on the third user device **4806**. The collection interface allows the third user to navigate the second content collection at block **4828**.

(336) The second public content collection may, for example, consist exclusively of combined user input content items created using a first user input content item as a base item and shared back to the first public content collection (e.g., public feed). Where the first user input content item and the supplemental content added to create the combined user input content item both include visual content (e.g., still images and/or video), the second public collection thus provides users with an automatic and convenient view of all visual responses to the base (original) item of the first user. Each combined user input content item in the second public content collection may include an attribution element which identifies the first user. In some examples, the collection interface may include an attribution visual element which identifies the first user or the base item, separately from a listing or display of the “remixed” items.

(337) FIG. **49** is a flowchart illustrating a method **4900**, according to some examples, to enable an active user to access a collection interface in a media application, e.g., the interaction client **106**. Accordingly, the method **4900** may be performed by components of the interaction system **100**. For ease of reference and to facilitate understanding, the four users and their respective user devices **4802**, **4804**, **4806** and **4808**, of FIG. **48**, are again referred to in the description provided with reference to FIG. **49** and FIG. **50**.

(338) The method **4900** commences at opening loop block **4902** and proceeds to block **4904**, where the media application causes presentation of a first public content collection, in the example form of a public feed, on the second user device **4804** of the second user. The public feed includes a number of user input content items that the second user can scroll through, react to, share, and the like. The second user wishes to react to one of these user input content items, being the base item, by creating a visual reaction thereto, and at block **4906**, the media application receives, from the second user device **4804** of the second user, an indication of a combination user input related to the base item. The indication of a combination user input related to the base item may, for example, be a selection of a combination graphical element by the second user. The media application may present a combination graphical element in conjunction with the base item on a user interface, with the combination graphical element being user-selectable to invoke the combination function (“remix” function) of the media application.

(339) The method **4900** then proceeds to block **4908** where, responsive to receiving the indication of the combination user input from the second user device **4804**, the media application invokes the combination function of the media application on the second user device **4804**. The combination function allows the second user to access a second user input content item and combine the first user input content item (base item) with the second user input content item to create a first combined user input content item.

(340) The remix system **206** of the interaction system **100** may detect a third-party authorization for the base item to be combined with other content items in order to create a combined user input content item. For example, an authorization setting related to the base item, indicating this authorization, may be detected. A content creator, e.g., the creator of the base item, when creating or publishing the item, may be presented with the option of either allowing or disallowing the use of the item in combination with other content items, created or originated with other users of the interaction system **100**. This allowance/disallowance is then recorded and stored in the database **128** as the authorization setting related to the item, and is accessed to perform the detection of the third-party authorization.

(341) A remix interface may be presented by the camera system **204** and specifically the remix system **206**, to enable a user to capture a further content item and combine the further content item with the base item viewed in the public feed, to generate a combined user input content item. The remix interface presents various layout tools for laying out multiple content items relative to each

other and within certain layouts. Various augmentation and modification tools may be applied to combined content within the context of the remix interface.

(342) Once a combined user input content item has been created, the second user may be presented with various sharing destinations by the media application. The second user opts to share the first combined user input content item to the public feed, and the interaction system **100** causes the first combined user input content item to become available in the public feed. Further, at block **4910**, in response to the sharing of the first combined user input content item to the public feed, the collection management system **226** causes the first combined user input content item to be stored in association with the base item in a second public content collection, as described above with reference to FIG. **48**. This allows users accessing the second public content collection to view the newly created “remix” of the base item.

(343) At block **4912**, the method **4900** includes receiving, from the third user device **4806** of the third user of the media application, an indication of a combination collection presentation user input related to the base item. This may include causing, by the media application on the third user device **4806**, presentation of a combination collection presentation graphical element in conjunction with the base item. The combination collection presentation graphical element is then user-selectable to invoke a combination collection presentation function.

(344) Responsive to receiving the indication of the combination collection presentation user input (e.g., user selection of the combination collection presentation graphical element), the combination collection presentation function of the media application is invoked on the third user device **4806** (block **4914**). The combination collection presentation function causes presentation of the collection interface as described with reference to FIG. **48**, and examples of which are further described below. The collection interface enables the third user to navigate the second public content collection at block **4916**, and may also automatically present a combination element to allow the third user to create their own visual response to the base item. The method **4900** concludes at closing loop block **4918**.

(345) FIG. **50** is a flowchart illustrating a method **5000**, according to some examples, to enable active users to access a collection interface in a media application, e.g., the interaction client **106**. The method **5000** may be performed by components of the interaction system **100**.

(346) The method **5000** starts at opening loop block **5002** and progresses to block **5004**, where the media application causes presentation of the public feed on the fourth user device **4808** of the fourth user. At block **5006**, the media application receives, from the fourth user device **4808** of the fourth user, an indication of a combination user input related to the base item. Similarly to the second user, the fourth user is then enabled to create a combined user input content item through the activation of the combination function at block **5008**.

(347) The method **5000** proceeds to block **5010**, where a third user input content item is accessed and combined with the base item to create a second combined user input content item. It will be appreciated that this process may include laying out the third user input content item relative to the first user input content item, as described elsewhere.

(348) At block **5012**, responsive to determining sharing, by the fourth user, of the second combined user input content item to the public feed, the interaction system **100** causes the second combined user input content item to be stored in association with the base item in the second public content collection. The interaction system **100** may access metadata of the second combined user input content item identifying it as a visual combinatory response to the first user input content item. Based on this identification, the interaction system **100** automatically adds the second combined user input content item to the second public content collection. In some examples, a combined user input content item is only added to the second public content collection if it was shared back to the public feed, or designated as “public” (or a similar designation), to ensure that combined user input content items intended for private use, or only intended for sharing with a user's connections or “friends,” are not added to the second public content collection for public navigation.

(349) Users of the interaction system **100**, such as the third user, can then access the second public content collection to view the first combined user input content item, the second combined user input content item, and any other visual combinatory responses to the base item (block **5014**). The method **5000** then ends at closing loop block **5016**.

(350) FIG. **51** to FIG. **54** are interface diagrams showing a sequence of user interfaces that may be generated and displayed by the interaction client **106**, according to some examples.

(351) FIG. **51** is a user interface diagram showing a viewing interface **5102**. An active (viewing) user, when scrolling through a public feed **5104**, can view a base item **5106**. The public feed **5104** is accessible via a dedicated public feed tab **5116** of the interaction client **106**. The base item **5106** is a video clip of a person dancing, with an audio clip (of the same duration) playing at the same time. In FIG. **51**, the base item **5106** was created and posted to the public feed **5104** by a first user, “John,” using a user device running an instance of the interaction client **106**. The first user, “John,” may for example have captured a person dancing to the tune of a third-party song—the third-party song is selectable to add it to the video clip as a feature of the interaction system **100**.

(352) The viewing interface **5102** includes a profile data element **5108** identifying the first user, and the active user may select the profile data element **5108** in order to view a profile of the first user. The viewing interface **5102** further includes an audio attribution data element **5110** (as an example of an audio identifier) identifying the audio clip and an augmentation identifier **5112** identifying an augmentation applied to the base item by the first user. Where the audio clip is a third-party song, the audio identifier may thus give suitable recognition to the artist associated with the song.

(353) Responsive to user selection of an action menu button **5114**, an action menu **5202** is presented.

(354) FIG. **52** is a user interface diagram showing the viewing interface **5102** with the action menu **5202** presented. One of the options presented to the active user by way of the action menu **5202** is a combination graphical element in the example form of a remix button **5204**. The remix button **5204** is user-selectable to invoke the media application's combination function. Selection of the remix button **5204** causes presentation of a remix interface **5302**.

(355) In some examples, the active user may perform a different action, such as a press and hold gesture on the base item **5106**, to invoke the combination function (with no action menu being presented).

(356) The remix interface **5302** (also referred to as a combination interface) is shown in FIG. **53**, according to some examples. The remix interface **5302** is presented by the camera system **204** and specifically the remix system **206**, responsive to user-selection of the remix button **5204**, to provide a modular camera enabling a user to capture a further content item and combine the further content item with the third-party content item (base item) viewed in the viewing interface **5102**, to generate a combined user input content item. The remix interface **5302** presents various augmentation and modification tools in an augmentation carousel **5304**, that may be applied to combined content within the context of the remix interface **5302**, and also presents various camera controls **5306**.

(357) While an augmentation carousel **5304** is shown in FIG. **53**, with a lens **5312** selected, in other examples, the remix interface **5302** may include only a remix capture button similar to the remix interface **4602** shown in FIG. **46** (without any lenses or augmentation elements).

(358) It will be evident from FIG. **53** that, in some examples, the user may capture the second user input content item such that it is overlaid over the base item. Specifically, in FIG. **53**, the active user captures a second user input content item **5308** in the example form of a video reaction to the video clip of the base item **5106**. However, various other layout options may be applied, as shown, for example, in the remix interface **4602** of FIG. **46**.

(359) The media application on the active user's device includes a camera function, causing presentation a digital image feed from at least one camera of the user device in the remix interface **5302**. The media application receives user capture input from the active user via the remix interface

**5302** and, in response thereto, captures digital image content (in FIG. **53**, this is in the form of a video clip) as the second user input content item **5308** and stores it to be accessible by the media application on the user device.

(360) Once captured, this creates a combined user input content item **5310**. In FIG. **53**, the combined user input content item **5310** is a combined video clip, showing both the video of the base item **5106** and the active user's video reaction/response. Further, the audio content associated with the base item **5106** is carried over into the combined user input content item **5310** such that the “original” audio clip from the base item **5106** also plays when the combined user input content item **5310** is played. Accordingly, the audio identifier of the base item **5106** can be carried through to the combined user input content item **5310** such that, when the combined user input content item **5310** is presented to another user, it is presented in conjunction with the audio identifier.

(361) After creating the combined user input content item **5310**, the active user may be enabled to save the combined user input content item **5310**, e.g., to a “Memories” tab, and may also share the combined user input content item **5310**, e.g., to the public feed of the interaction system **100**.

(362) FIG. **54** is a user interface diagram showing a bottom region of the finalized and published combined user input content item **5310**, being viewed by a third-party user in a viewing interface **5402**. The third-party user views the combined user input content item **5310** via the public feed tab **5116** providing access to the public feed.

(363) The viewing interface **5402** includes a profile data element **5404** identifying the user responsible for creating the “remix” (in this case, “Sally”), as well as a base item attribution element **5408**. The attribution element **5408** identifies the creator of the base item (“first user”, in this case, “John”), being the original content that was “remixed.” In some examples, the base item attribution element **5408** may identify the base item itself. The elements **5404** and **5408** are user-selectable to enable the third-party user to view the profile of the user associated with the element **5404/5408**. In some examples, the base item attribution element **5408** is user-selectable to cause presentation of the (original) base item. In other words, the third-party user can select the base item attribution element **5408** to view the base item without any combination content.

(364) The viewing interface **5402** further includes an audio attribution data element **5110** (as an example of an audio identifier) identifying the audio clip. This is the same audio clip as the audio clip identified in FIG. **51**. As mentioned above, in examples where a third-party song is used, the audio identifier gives the appropriate credit to the artist. An augmentation identifier **5406** identifies an augmentation applied to the combined user input content item.

(365) The interaction system **100** may be configured to handle attribution data differently for users with different profile types. For example, if a user has a public profile, an attribution element like the base item attribution element **5408** may be shown, which includes the identity of the creator. If a user has a private profile, the attribution element may simply indicate, for example, that the content item has been “remixed from the public feed,” without revealing the identity of the creator.

(366) As alluded to above, an active user may wish to view combined user input content items (“remixes”) made from a particular base item in a consolidated location. FIG. **55** and FIG. **56** are user interface diagrams showing different entry points to a collection interface that provides such a view, together with further functionality, according to some examples.

(367) In FIG. **55**, a viewing interface **5502** displays a base item **5504** being viewed on the public feed **5104**. The viewing interface **5502** presents, to the active user, a combination collection presentation graphical element **5506**. The combination collection presentation graphical element **5506** is shown in conjunction with the base item **5504**, and is user-selectable to invoke a combination collection presentation function. In other words, responsive to receiving combination collection presentation user input in the example form of user selection of the combination collection presentation graphical element **5506**, the combination collection presentation function is invoked.

(368) In some examples, the interaction system **100** may thus determine that a base item has been

used to create one or more combined user input content items that were shared publicly in the interaction system **100**, and in response, cause the combination collection presentation graphical element **5506** to be presented automatically in the viewing interface **5502**. If a base item was not shared publicly, e.g., was only shared as a direct message to a specific user or to the sharing user's private story, no combination collection presentation graphical element is presented.

(369) In FIG. **56**, a viewing interface **5602** does not display a combination collection presentation graphical element automatically. Instead, in response to user selection of an action menu button (similar to the action menu button **5114**) while viewing a base item **5604**, an action menu **5608** is presented. The action menu **5608** includes a combination collection presentation graphical element **5606**.

(370) FIG. **57** is a user interface diagram showing a collection interface **5702**, according to some examples. The collection interface **5702** is presented in response to user selection of the combination collection presentation graphical element **5606**, and allows the viewing user to navigate all combined user input content items created from the base item **5604**.

(371) The collection interface **5702** includes, in a top section, a base item thumbnail **5704** identifying the base item **5604**, together with a written base item identifier **5706**. Combination data **5710**, in the case of FIG. **57** indicating that the base item **5604** has been “remixed” a total of 106 times, is also displayed in the top section.

(372) Below the top section, the collection interface **5702** presents a scrollable (or otherwise navigable) grid **5708** of all “remixes” created from the base item **5604**. In the case of FIG. **57**, the viewing user can scroll through all the “remixes,” select them, react to them, etc. This collection of “remixes,” all linked to the same base item, is referred to in some examples as a “second public content collection” (the general public feed being the “first public content collection” in some examples).

(373) The collection interface **5702** presents a remix button **5712** to allow the viewing user to create a further “remix” of the base item **5604** and, if the user opts to do so, add it to the second public content collection. Accordingly, in some examples, activating the combination collection presentation function of the media application on a user device (e.g., user system **104**) may include causing presentation of a combination graphical element, e.g., a remix button, within the collection interface, with the combination graphical element being user-selectable to invoke the combination function to allow the user to access (e.g., capture) a further user input content item and combine the original (base) user input content item with the further user input content item to create another combined user input content item. This provides a convenient way for a viewing user not only to view all “remixes” linked to a certain original content item, but also to automatically have the option to create their own “remix.” A “remix” may be seen as a way of responding, reacting or replying to another user's post, and examples of the present disclosure thus provide a technological tool facilitating community creativity, allowing users to express themselves and interact across such a community. In some examples, e.g., in the case of FIG. **57**, the “remixes” are visual combinatory responses to a creator's original/base item, and each user creating a combined user input content item can add new visual content in response to the original/base item.

(374) FIG. **58** is a user interface diagram showing a viewing interface **5802**, according to some examples, presenting a view of a base item from which combined user input content items have been made. The viewing interface **5802** is presented to the creator of the base item. In other words, when a user is viewing their own user input content item that has been posted to the public feed, an interface such as the viewing interface **5802** may be presented.

(375) The viewing interface **5802** includes a combination information cell **5804**. In some examples, the combination information cell **5804** displays a count of all “remixes” made by users from the base item. This count may include all “remixes,” irrespective of whether they were shared publicly. In other words, private content such as a “remix” shared only via direct message may also be added to the count. This provides a creator with an indication of the degree of interaction with the base

item by users, together with other counts, such as the number of likes (see the element **5806**) and the number of shares (see the element **5808**).

(376) In some examples, the combination information cell **5804** is user-selectable to invoke the combination collection presentation function. In other words, responsive to receiving combination collection presentation user input in the example form of user selection of the combination information cell **5804**, the combination collection presentation function is invoked to present, to the creator of the base item, a collection interface. The collection interface, however, only includes “remixes” that have been publicly shared or posted, hence the number of combined user input content items in the second public content collection may be less than the count shown to the creator in FIG. **58**.

(377) In order to inform users and facilitate use of the aforementioned features, a tooltip may be displayed to users to alert them of the option to create a combined user input content item from a base item being viewed in the public feed. FIG. **59** is a user interface diagram showing a viewing interface **5902**, according to some examples, presenting a view of a base item in the public feed tab **5116**. The action menu button **5114** is presented together with a tooltip **5904** pointing to the action menu button **5114**. This alerts the user to the option available in the action menu, as described with reference to FIG. **52**.

(378) Data Communications Architecture

(379) FIG. **60** is a schematic diagram illustrating a structure of a message **6000**, according to some examples, generated by an interaction client **106** for communication to a further interaction client **106** via the interaction servers **124**. The content of a particular message **6000** is used to populate the message table **306** stored within the database **304**, accessible by the interaction servers **124**. Similarly, the content of a message **6000** is stored in memory as “in-transit” or “in-flight” data of the user system **104** or the interaction servers **124**. A message **6000** is shown to include the following example components: Message identifier **6002**: a unique identifier that identifies the message **6000**. Message text payload **404**: text, to be generated by a user via a user interface of the user system **104**, and that is included in the message **6000**. Message image payload **6004**: image data, captured by a camera component of a user system **104** or retrieved from a memory component of a user system **104**, and that is included in the message **6000**. Image data for a sent or received message **6000** may be stored in the image table **6006**. Message video payload **6008**: video data, captured by a camera component or retrieved from a memory component of the user system **104**, and that is included in the message **6000**. Video data for a sent or received message **6000** may be stored in the video table **6010**. Message audio payload **6012**: audio data, captured by a microphone or retrieved from a memory component of the user system **104**, and that is included in the message **6000**. Message augmentation data **6014**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **6004**, message video payload **6008**, or message audio payload **6012** of the message **6000**. Augmentation data for a sent or received message **6000** may be stored in the augmentation table **6016**. Message duration parameter **6018**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **6004**, message video payload **6008**, message audio payload **6012**) is to be presented or made accessible to a user via the interaction client **106**. Message geolocation parameter **6020**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **6020** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload **6004**, or a specific video in the message video payload **6008**). Message story identifier **6022**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **6024**) with which a particular content item in the message image payload **6004** of the message **6000** is associated. For example, multiple images within the message image payload **6004** may each be associated with multiple content collections using identifier

values. Message tag **6026**: each message **6000** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **6004** depicts an animal (e.g., a lion), a tag value may be included within the message tag **6026** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition. Message sender identifier **6028**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system **104** on which the message **6000** was generated and from which the message **6000** was sent. Message receiver identifier **6030**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system **104** to which the message **6000** is addressed. (380) The contents (e.g., values) of the various components of message **6000** may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload **6004** may be a pointer to (or address of) a location within an image table **6006**. Similarly, values within the message video payload **6008** may point to data stored within a video table **6010**, values stored within the message augmentation data **6014** may point to data stored in an augmentation table **6016**, values stored within the message story identifier **6022** may point to data stored in a story table **6024**, and values stored within the message sender identifier **6028** and the message receiver identifier **6030** may point to user records stored within an entity table **6032**.

(381) Machine Architecture

(382) FIG. **61** is a diagrammatic representation of the machine **6100** within which instructions **6102** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **6100** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **6102** may cause the machine **6100** to execute any one or more of the methods described herein. The instructions **6102** transform the general, non-programmed machine **6100** into a particular machine **6100** programmed to carry out the described and illustrated functions in the manner described. The machine **6100** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **6100** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **6100** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **6102**, sequentially or otherwise, that specify actions to be taken by the machine **6100**. Further, while a single machine **6100** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **6102** to perform any one or more of the methodologies discussed herein. The machine **6100**, for example, may comprise the user system **104** or any one of multiple server devices forming part of the interaction server system **102**. In some examples, the machine **6100** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

(383) The machine **6100** may include processors **6104**, memory **6106**, and input/output I/O components **6108**, which may be configured to communicate with each other via a bus **6110**. In an example, the processors **6104** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable

combination thereof) may include, for example, a processor **6112** and a processor **6114** that execute the instructions **6102**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **61** shows multiple processors **6104**, the machine **6100** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

(384) The memory **6106** includes a main memory **6116**, a static memory **6118**, and a storage unit **6120**, both accessible to the processors **6104** via the bus **6110**. The main memory **6106**, the static memory **6118**, and storage unit **6120** store the instructions **6102** embodying any one or more of the methodologies or functions described herein. The instructions **6102** may also reside, completely or partially, within the main memory **6116**, within the static memory **6118**, within machine-readable medium **6122** within the storage unit **6120**, within at least one of the processors **6104** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **6100**.

(385) The I/O components **6108** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **6108** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **6108** may include many other components that are not shown in FIG. **61**. In various examples, the I/O components **6108** may include user output components **6124** and user input components **6126**. The user output components **6124** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **6126** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(386) In further examples, the I/O components **6108** may include biometric components **6128**, motion components **6130**, environmental components **6132**, or position components **6134**, among a wide array of other components. For example, the biometric components **6128** include components to determine expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **6130** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

(387) The environmental components **6132** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that determine ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that determine background noise), proximity sensor components (e.g., infrared sensors that determine nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants



in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

(388) With respect to cameras, the user system **104** may have a camera system comprising, for example, front cameras on a front surface of the user system **104** and rear cameras on a rear surface of the user system **104**. The front cameras may, for example, be used to capture still images and video of a user of the user system **104** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the user system **104** may also include a 360° camera for capturing 360° photographs and videos.

(389) Further, the camera system of the user system **104** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the user system **104**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

(390) The position components **6134** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that determine air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(391) Communication may be implemented using a wide variety of technologies. The I/O components **6108** further include communication components **6136** operable to couple the machine **6100** to a network **6138** or devices **6140** via respective coupling or connections. For example, the communication components **6136** may include a network interface component or another suitable device to interface with the network **6138**. In further examples, the communication components **6136** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **6140** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

(392) Moreover, the communication components **6136** may determine identifiers or include components operable to determine identifiers. For example, the communication components **6136** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to determine one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **6136**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via determining an NFC beacon signal that may indicate a particular location, and so forth.

(393) The various memories (e.g., main memory **6116**, static memory **6118**, and memory of the processors **6104**) and storage unit **6120** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **6102**), when executed by processors **6104**, cause various operations to implement the disclosed examples.

(394) The instructions **6102** may be transmitted or received over the network **6138**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **6136**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **6102** may be

transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **6140**.

(395) System with Head-Wearable Apparatus

(396) FIG. **62** illustrates a user system **104** including a head-wearable apparatus **116** with a selector input device, according to some examples. FIG. **62** is a high-level functional block diagram of an example head-wearable apparatus **116** communicatively coupled to a mobile device **114** and a server system **6204** via various network **6206**.

(397) The head-wearable apparatus **116** includes one or more cameras, each of which may be, for example, a visible light camera **6208**, an infrared emitter **6210**, and an infrared camera **6212**.

(398) The mobile device **114** connects with head-wearable apparatus **116** using both a low-power wireless connection **6214** and a high-speed wireless connection **6216**. The mobile device **114** is also connected to the server system **6204** and the network **6206**.

(399) The head-wearable apparatus **116** further includes two image displays of the image display of optical assembly **6218**. The two image displays of optical assembly **6218** include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus **116**. The head-wearable apparatus **116** also includes an image display driver **6220**, an image processor **6222**, low-power circuitry **6224**, and high-speed circuitry **6226**. The image display of optical assembly **6218** is for presenting images and videos, including an image that can include a graphical user interface to a user of the head-wearable apparatus **116**.

(400) The image display driver **6220** commands and controls the image display of optical assembly **6218**. The image display driver **6220** may deliver image data directly to the image display of optical assembly **6218** for presentation or may convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H.264 (MPEG-4 Part 10), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to compression formats such as Portable Network Group (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (EXIF) or the like.

(401) The head-wearable apparatus **116** includes a frame and stems (or temples) extending from a lateral side of the frame. The head-wearable apparatus **116** further includes a user input device **6228** (e.g., a touch sensor or push button), including an input surface on the head-wearable apparatus **116**. The user input device **6228** (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

(402) The components shown in FIG. **62** for the head-wearable apparatus **116** are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively, or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus **116**. Left and right visible light cameras **6208** can include digital camera elements such as a complementary metal oxide—semiconductor (CMOS) image sensor, charge-coupled device, camera lenses, or any other respective visible or light-capturing elements that may be used to capture data, including images of scenes with unknown objects.

(403) The head-wearable apparatus **116** includes a memory **6202**, which stores instructions to perform a subset or all of the functions described herein. The memory **6202** can also include storage device.

(404) As shown in FIG. **62**, the high-speed circuitry **6226** includes a high-speed processor **6230**, a memory **6202**, and high-speed wireless circuitry **6232**. In some examples, the image display driver **6220** is coupled to the high-speed circuitry **6226** and operated by the high-speed processor **6230** in order to drive the left and right image displays of the image display of optical assembly **6218**. The high-speed processor **6230** may be any processor capable of managing high-speed communications and operation of any general computing system needed for the head-wearable apparatus **116**. The high-speed processor **6230** includes processing resources needed for managing high-speed data transfers on a high-speed wireless connection **6216** to a wireless local area network (WLAN) using

the high-speed wireless circuitry **6232**. In certain examples, the high-speed processor **6230** executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus **116**, and the operating system is stored in the memory **6202** for execution. In addition to any other responsibilities, the high-speed processor **6230** executing a software architecture for the head-wearable apparatus **116** is used to manage data transfers with high-speed wireless circuitry **6232**. In certain examples, the high-speed wireless circuitry **6232** is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as WiFi. In some examples, other high-speed communications standards may be implemented by the high-speed wireless circuitry **6232**.

(405) The low-power wireless circuitry **6234** and the high-speed wireless circuitry **6232** of the head-wearable apparatus **116** can include short-range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). mobile device **114**, including the transceivers communicating via the low-power wireless connection **6214** and the high-speed wireless connection **6216**, may be implemented using details of the architecture of the head-wearable apparatus **116**, as can other elements of the network **6206**.

(406) The memory **6202** includes any storage device capable of storing various data and applications, including, among other things, camera data generated by the left and right visible light cameras **6208**, the infrared camera **6212**, and the image processor **6222**, as well as images generated for display by the image display driver **6220** on the image displays of the image display of optical assembly **6218**. While the memory **6202** is shown as integrated with high-speed circuitry **6226**, in some examples, the memory **6202** may be an independent standalone element of the head-wearable apparatus **116**. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor **6230** from the image processor **6222** or the low-power processor **6236** to the memory **6202**. In some examples, the high-speed processor **6230** may manage addressing of the memory **6202** such that the low-power processor **6236** will boot the high-speed processor **6230** any time that a read or write operation involving memory **6202** is needed.

(407) As shown in FIG. **62**, the low-power processor **6236** or high-speed processor **6230** of the head-wearable apparatus **116** can be coupled to the camera (visible light camera **6208**, infrared emitter **6210**, or infrared camera **6212**), the image display driver **6220**, the user input device **6228** (e.g., touch sensor or push button), and the memory **6202**.

(408) The head-wearable apparatus **116** is connected with a host computer. For example, the head-wearable apparatus **116** is paired with the mobile device **114** via the high-speed wireless connection **6216** or connected to the server system **6204** via the network **6206**. The server system **6204** may be one or more computing devices as part of a service or network computing system, for example, that includes a processor, a memory, and network communication interface to communicate over the network **6206** with the mobile device **114** and the head-wearable apparatus **116**.

(409) The mobile device **114** includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network **6206**, low-power wireless connection **6214**, or high-speed wireless connection **6216**. mobile device **114** can further store at least portions of the instructions for generating binaural audio content in the mobile device **114**'s memory to implement the functionality described herein.

(410) Output components of the head-wearable apparatus **116** include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light-emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver **6220**. The output components of the head-wearable apparatus **116** further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus **116**, the mobile device **114**, and server system **6204**, such as the user input device **6228**, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive

alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(411) The head-wearable apparatus **116** may also include additional peripheral device elements. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with the head-wearable apparatus **116**. For example, peripheral device elements may include any I/O components including output components, motion components, position components, or any other such elements described herein.

(412) For example, the biometric components include components to determine expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), Wi-Fi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that determine air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over low-power wireless connections **6214** and high-speed wireless connection **6216** from the mobile device **114** via the low-power wireless circuitry **6234** or high-speed wireless circuitry **6232**.

(413) Software Architecture

(414) FIG. **63** is a block diagram **6300** illustrating a software architecture **6302**, which can be installed on any one or more of the devices described herein. The software architecture **6302** is supported by hardware such as a machine **6304** that includes processors **6306**, memory **6308**, and I/O components **6310**. In this example, the software architecture **6302** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **6302** includes layers such as an operating system **6312**, libraries **6314**, frameworks **6316**, and applications **6318**. Operationally, the applications **6318** invoke API calls **6320** through the software stack and receive messages **6322** in response to the API calls **6320**.

(415) The operating system **6312** manages hardware resources and provides common services. The operating system **6312** includes, for example, a kernel **6324**, services **6326**, and drivers **6328**. The kernel **6324** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **6324** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services **6326** can provide other common services for the other software layers. The drivers **6328** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **6328** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

(416) The libraries **6314** provide a common low-level infrastructure used by the applications **6318**. The libraries **6314** can include system libraries **6330** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **6314** can include API libraries **6332** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate

(AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **6314** can also include a wide variety of other libraries **6334** to provide many other APIs to the applications **6318**.

(417) The frameworks **6316** provide a common high-level infrastructure that is used by the applications **6318**. For example, the frameworks **6316** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **6316** can provide a broad spectrum of other APIs that can be used by the applications **6318**, some of which may be specific to a particular operating system or platform.

(418) In an example, the applications **6318** may include a home application **6336**, a contacts application **6338**, a browser application **6340**, a book reader application **6342**, a location application **6344**, a media application **6346**, a messaging application **6348**, a game application **6350**, and a broad assortment of other applications such as a third-party application **6352**. The applications **6318** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **6318**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **6352** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **6352** can invoke the API calls **6320** provided by the operating system **6312** to facilitate functionalities described herein.

## CONCLUSION

(419) Although the examples, including exemplary flow diagrams, may depict a particular sequence of operations, sequence/s may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the described component, system or process. In other examples, different components of an example device or system that implements a method or process, may perform functions at substantially the same time or in a specific sequence.

(420) As used in this disclosure, phrases of the form “at least one of an A, a B, or a C,” “at least one of A, B, or C,” “at least one of A, B, and C,” and the like, should be interpreted to select at least one from the group that comprises “A, B, and C.” Unless explicitly stated otherwise in connection with a particular instance in this disclosure, this manner of phrasing does not mean “at least one of A, at least one of B, and at least one of C.” As used in this disclosure, the example “at least one of an A, a B, or a C,” would cover any of the following selections: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, and {A, B, C}.

(421) The terms “combined” and “combination”, e.g., when used to refer to combined content items and/or combination functions, refer to content items or content data that have been, or may be, combined in any suitable manner. Examples may include combinations of two or more images, combinations of two or more video clips, or the like. Examples may also include combinations of different content types, e.g., a video clip combined with a still image, or a video clip combined with an audio item, or combinations thereof. Such combinations may be arranged in different ways, e.g., items may be displayed or played concurrently, sequentially, or in an overlapping fashion, and items may be arranged using different layout options, e.g., overlays, split screens, or the like, with examples being described herein.

(422) “Carrier signal” refers to any intangible medium that is capable of storing, encoding, or

carrying instructions for execution by the machine and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

(423) “Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network. In a context where such a device is used by a user, a “client device” may be referred to as a “user device”.

(424) “Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth-generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

(425) “Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processors. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to

perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

(426) “Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-

readable medium” mean the same thing and may be used interchangeably in this disclosure.

(427) “Ephemeral message” refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

(428) “Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

(429) “Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

(430) “Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

(431) “User input content” refers to any form of content that may be posted, published, shared, sent, received or interacted with by a user, e.g., using an online platform. User input content may, for example, include digital content in the form of one or more image, photo, video clip, audio clip, or a combination thereof. A user input content item may, in some examples, be a UGC (user-generated content) item. User input content may, for example, have an augmentation applied to it (e.g., a lens, a filter, a media overlay, or other augmentation). A user input content item may be generated by a user, for example, where the content item is captured by a user using a digital camera of a user device, or, for example, where the content item is created by the user by applying an augmentation to another content item. A user input content item may be user-selected, for example, where the content item is selected from a media library or file storage, or, for example, sourced from a third party or third-party device. User input content may also include partially or fully computer-generated content, e.g., content created through generative artificial intelligence. A content item may, for example, be content data, such as a picture, video or audio data (or combinations thereof).

## Claims

1. A method to facilitate unlocking of a content sharing destination, the method comprising: causing presentation of a third-party user input content item on a user system of a user; determining user selection of a content sharing graphical element; responsive to the determining of the user selection of the content sharing graphical element, invoking a content sharing function, the



invoking of the content sharing function comprising causing presentation of a destination graphical element on the user system, the destination graphical element identifying a first content sharing destination, and the first content sharing destination being locked; causing presentation of a combination graphical element, the combination graphical element being user-selectable to invoke a combination function based on a third-party authorization for the third-party user input content item to be combined with other content items; determining user selection of the combination graphical element; and responsive to the determining of the user selection of the combination graphical element, invoking the combination function to: access a second user input content item; combine the third-party user input content item with the second user input content item to create a combined user input content item; unlock the first content sharing destination; and enable the user to share the combined user input content item to the unlocked first content sharing destination.

2. The method of claim 1, wherein causing the presentation of the third-party user input content item on the user system of the user comprises causing presentation of the third-party user input content item within a viewing interface in conjunction with the content sharing graphical element, the content sharing graphical element being user-selectable to invoke the content sharing function.

3. The method of claim 2, wherein the invoking of the content sharing function comprises causing presentation of a plurality of destination graphical elements on the user system, each destination graphical element identifying a content sharing destination, and the content sharing destinations comprising the first content sharing destination and one or more additional content sharing destinations.

4. The method of claim 3, wherein the first content sharing destination is locked for the third-party user input content item and the one or more additional content sharing destinations are unlocked for the third-party user input content item, the destination graphical element identifying each respective additional content sharing destination being user-selectable to enable the user to share the third-party user input content item to the additional content sharing destination associated with the destination graphical element.

5. The method of claim 1, wherein the first content sharing destination is a content collection associated with the user.

6. The method of claim 5, wherein the content collection associated with the user is a story, the story being created and published by the user via a media application to be accessible by a plurality of other users of the media application.

7. The method of claim 1, wherein the destination graphical element is presented within a sharing interface, the sharing interface being presented responsive to determining the user selection of the content sharing graphical element.

8. The method of claim 1, wherein enabling the user to share the combined user input content item to the unlocked first content sharing destination comprises: causing presentation of an updated destination graphical element on the user system, the updated destination graphical element identifying the unlocked first content sharing destination; determining user selection of the updated destination graphical element; and responsive to determining the user selection of the updated destination graphical element, causing the combined user input content item to be published to the unlocked first content sharing destination.

9. The method of claim 8, wherein causing the presentation of the updated destination graphical element on the user system comprises causing the updated destination graphical element to be presented within a sharing interface.

10. The method of claim 1, wherein causing the presentation of the combination graphical element comprises causing the combination graphical element to be presented within or adjacent to the destination graphical element identifying the first content sharing destination.

11. The method of claim 1, comprising detecting a sharing setting associated with the third-party user input content item, the sharing setting indicating that the third-party user input content item is not directly shareable to the first content sharing destination.

12. The method of claim 11, wherein the sharing setting indicates that the third-party user input content item is shareable to the first content sharing destination as part of combined user input content items.

13. The method of claim 1, wherein the combination function comprises a camera function that enables the user to capture the second user input content item.

14. The method of claim 1, wherein the method further comprises: causing presentation of a digital image feed from at least one camera of the user system; receiving a user capture input; responsive to receipt of the user capture input, capturing digital content as the second user input content item; and storing the second user input content item.

15. The method of claim 1, wherein the combination function comprises a camera function that enables the user to select the second user input content item from a storage component.

16. The method of claim 1, wherein the combination function comprises a layout function to enable the user to specify a layout relationship of the third-party user input content item relative to the second user input content item during presentation of the combined user input content item.

17. The method of claim 1, wherein the third-party user input content item is part of a set of third-party user input content items created by one or more third-party user of a media application, and the presentation of the third-party user input content item is part of a sequential presentation of the set of third-party user input content items.

18. The method of claim 1, comprising causing display of a tooltip in conjunction with the combination graphical element, the tooltip comprising a message indicating that the first content sharing destination is unlockable via the combination function.

19. A computing apparatus comprising: a processor; and a memory storing instructions that, when executed by the processor, configure the apparatus to perform operations comprising: causing presentation of a third-party user input content item on a user system of a user; determining user selection of a content sharing graphical element; responsive to the determining of the user selection of the content sharing graphical element, invoking a content sharing function, the invoking of the content sharing function comprising causing presentation of a destination graphical element on the user system, the destination graphical element identifying a first content sharing destination, and the first content sharing destination being locked; causing presentation of a combination graphical element, the combination graphical element being user-selectable to invoke a combination function based on a third-party authorization for the third-party user input content item to be combined with other content items; determining user selection of the combination graphical element; and responsive to the determining of the user selection of the combination graphical element, invoking the combination function to: access a second user input content item; combine the third-party user input content item with the second user input content item to create a combined user input content item; unlock the first content sharing destination; and enable the user to share the combined user input content item to the unlocked first content sharing destination.

20. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising: causing presentation of a third-party user input content item on a user system of a user; determining user selection of a content sharing graphical element; responsive to the determining of the user selection of the content sharing graphical element, invoking a content sharing function, the invoking of the content sharing function comprising causing presentation of a destination graphical element on the user system, the destination graphical element identifying a first content sharing destination, and the first content sharing destination being locked; causing presentation of a combination graphical element, the combination graphical element being user-selectable to invoke a combination function based on a third-party authorization for the third-party user input content item to be combined with other content items; determining user selection of the combination graphical element; and responsive to the determining of the user selection of the combination graphical element, invoking the combination function to: access a second user input content item;

combine the third-party user input content item with the second user input content item to create a combined user input content item; unlock the first content sharing destination; and enable the user to share the combined user input content item to the unlocked first content sharing destination.

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